

Lecture 8: Forecasting Models
Linear Models, NVAR and Neural Networks
Neural ODEs, PINNs, Hybrid Models

Deep Learning in Classical and Quantum Physics

Marcin Płodzień

Atomic Optics Department
Institute of Theoretical Physics
Jagiellonian University in Kraków

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Recap



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The Law of Motion F

$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$: everything that drives the system

F contains all effects acting on the state: forces, couplings between the components, friction and dissipation, external driving, delays, and all parameters.

F known well

pendulum, springs, electrical circuits, planetary orbits

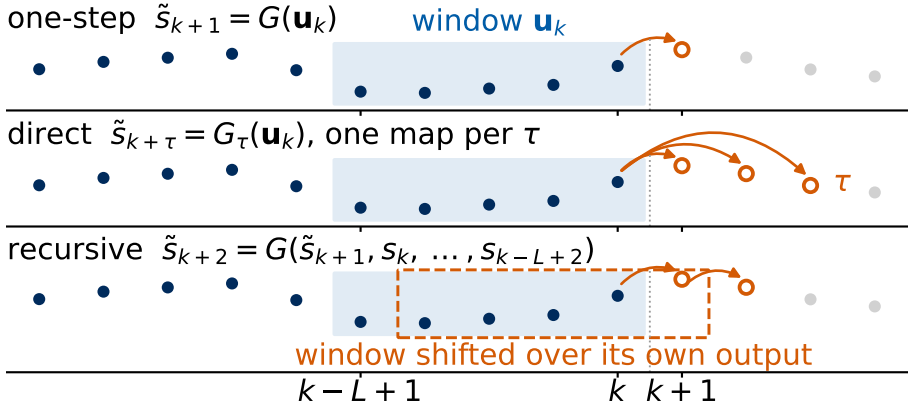
F known only approximately

cells, gene regulation, ecosystems, climate, the brain

missing terms, uncertain parameters, unknown couplings

- where **F** is imperfect, the record is the main source of knowledge: learn **F** from data, or combine the known part of the law with data

Forecast Task



filled dots: record; shaded: window $\mathbf{u}_k$; grey: future values; open circles: forecasts; top: one-step; middle: direct; bottom: recursive, the window (dashed) shifts over the outputs; model $\tilde{G}_\tau$, written G in the drawing

Forecast Map and Capacity

Forecast at horizon τ

$$\tilde{s}_{k+\tau} = \tilde{G}_\tau(\mathbf{u}_k)$$
$$\mathbf{u}_k = (s_k, \ldots, s_{k-L+1})$$

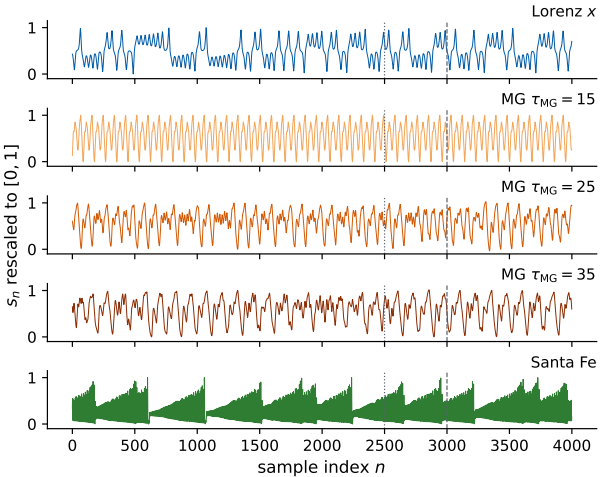
$\tilde{G}_\tau$: model for the value τ steps ahead; $\mathbf{u}_k$: window of the last $L \geq (m-1)\kappa + 1$ values; m: embedding dimension, κ: delay in samples

Capacity

$$C(\tau) = r^2\left(\tilde{s}_{k+\tau}, s_{k+\tau}\right)$$
$$C_{\text{tot}} = \sum_{\tau=1}^{50} C(\tau)$$

r: Pearson correlation over the test origins; $C_{\text{tot}} \leq 50$

Five Signals



Lorenz (1963); Mackey, Glass (1977); Hübner *et al.* (1989)

rescaled signal s_k against the sample index k, written s_n and n in the drawing; dotted: start of validation, dashed: start of the test block

- Lorenz-63, coordinate x
- Mackey–Glass (MG), delay $\tau_{\mathrm{MG}} = 15$ (limit cycle), 25 and 35 (chaos)
- Santa Fe: measured intensity of a chaotic laser

Characteristics of the Signals

signal	κ	m	$L = (m-1)\kappa + 1$	D_{KY}	λ_1 per sample	T_λ (samples)
Lorenz x	10	4	31	2.06	0.0181	55
MG 15	8	2	9	1.00	≈ 0	∞
MG 25	15	4	46	3.16	0.0099	101
MG 35	21	4	64	3.41	0.0029	340
Santa Fe	2	3	5	2.05 / 2.16	0.029	34

κ: first minimum of the mutual information (MI); m: false nearest neighbours (FNN); λ_1, D_{KY}: from the equations, for the laser from the record ($m = 3\,/\,4$); $T_\lambda = 1/\lambda_1$; κ, L, T_λ in samples

- FNN gives $m \leq 2D_{\mathrm{KY}}$, the estimate of the sufficient bound $2D_B$; MI and FNN: a heuristic window, models read longer ones
- $\tau \leq 50$: about one T_λ for Lorenz, within one for MG 25, 35; laser $T_\lambda = 34$ or 15 samples (two estimators), so 50 samples are 1.5–3.3 T_λ

Delay Window and the Forecast Map

known: the record $s_1, \ldots, s_k$ only; unknown: the law $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$, the state $\mathbf{x}_k$, the measurement function h in $s_k = h(\mathbf{x}_k)$

Delay vector and the embedding theorem

$$\mathcal{E}(\mathbf{x}_k) = (s_k, s_{k-\kappa}, \ldots, s_{k-(m-1)\kappa})$$

$\mathcal{E}$: delay map; $m > 2D_B$ (box-counting dimension): $\mathcal{E}$ one-to-one on the attractor for almost every smooth h, a sufficient condition; in practice m from FNN

Forecast map

$$G_\tau = h \circ \varphi_{\tau\Delta t} \circ \mathcal{E}^{-1}, \qquad G_\tau(\mathcal{E}(\mathbf{x}_k)) = s_{k+\tau} \text{ (noise-free)}$$

$\varphi_{\tau\Delta t}$: flow map over τ samples; G_τ exists, unknown, estimated by a model $\tilde{G}_\tau$; chaos limits its horizon to $T_{\text{pred}} = T_\lambda \ln(\Delta/\delta_0)$, δ_0: initial error, Δ: tolerated error

F. Takens, Lect. Notes Math. 898, 366 (1981); T. Sauer, J. A. Yorke, M. Casdagli, J. Stat. Phys. 65, 579 (1991)

Physics Knowledge in the Model

model	known	learned
AR, MA, NVAR	only the record s_k	linear readout of fixed features of $\mathbf{u}_k$
delay DMD (Koopman)	only the record s_k	linear map of the window
MLP, RNN, GRU, LSTM	only the record s_k	features and readout
Neural ODE	form $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$, measured state	vector field $\mathbf{F}_\theta$
PINN	the whole law	solution $x_\theta(t)$, unknown parameters
hybrid model	part of $\mathbf{F}$	missing term $\mathbf{U}_\theta$, then a formula

AR: autoregression; MA: moving average; NVAR: nonlinear vector autoregression; DMD: dynamic mode decomposition; MLP: multilayer perceptron; RNN: recurrent neural network; GRU: gated recurrent unit; LSTM: long short-term memory; ODE: ordinary differential equation; PINN: physics-informed neural network

- first models that see only the record, then models with physics inside
- each method next to a standard method for the same task

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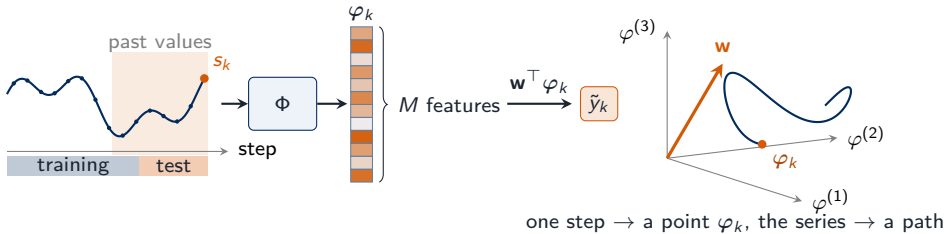
Linear Models



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Features and Linear Readout

$$(s_1, \ldots, s_k) \xrightarrow{\Phi} \varphi_k \in \mathbb{R}^M \xrightarrow{\text{linear}} \tilde{y}_k = \mathbf{w}^\top \varphi_k$$



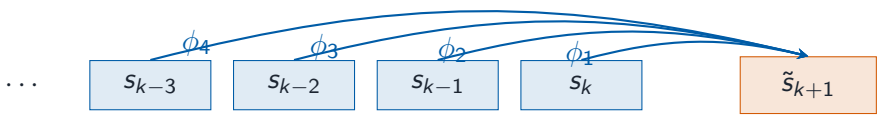
- Φ: M features of the last L values or a state $\mathbf{h}_k$ updated from $\mathbf{h}_{k-1}$ and s_k; linear (the past values) or nonlinear (products, tanh units)
- autoregression (AR) and nonlinear vector autoregression (NVAR): Φ fixed, only the readout $\mathbf{w} \in \mathbb{R}^M$ trained, by least squares
- neural networks train Φ and $\mathbf{w}$ together

Autoregressive Model AR(p)

AR(p): weighted sum of the last p values

$$\tilde{s}_{k+1} = \phi_0 + \sum_{i=1}^{p} \phi_i \, s_{k+1-i}$$

ϕ_0: constant; ϕ_i: weight of the value $i-1$ steps before s_k; p: order; record model $s_k = \phi_0 + \sum_i \phi_i s_{k-i} + \varepsilon_k$, innovation ε_k: white noise



- linear in the parameters: one least-squares problem

G. E. P. Box, G. M. Jenkins, G. C. Reinsel, G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed., Wiley (2015)

Fitting an AR Model

The parameters enter linearly, so fitting is one least-squares problem. Each row of X is one window of past values, and $\mathbf{y}$ holds the value that followed it.

$$X = \begin{pmatrix} 1 & s_{p-1} & \cdots & s_0 \\ \vdots & \vdots & & \vdots \\ 1 & s_{n-2} & \cdots & s_{n-p-1} \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} s_p \\ \vdots \\ s_{n-1} \end{pmatrix}, \quad \begin{pmatrix} \tilde{\phi}_0 \\ \tilde{\phi}_1 \\ \vdots \\ \tilde{\phi}_p \end{pmatrix} = \arg\min_{\phi} \|\mathbf{y} - X\phi\|^2$$

- no gradient descent is needed, the solution follows from the normal equations $X^\top X\tilde{\phi} = X^\top \mathbf{y}$
- our JAX implementation agrees with `statsmodels` to round-off

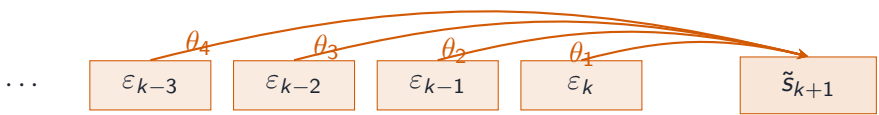
S. Seabold, J. Perktold, Proc. 9th Python in Science Conf., 92 (2010), doi:10.25080/Majora-92bf1922-011

Moving-Average Model MA(q)

MA(q): mean plus the last q innovations

$$s_k = \mu + \varepsilon_k + \sum_{i=1}^{q} \theta_i \, \varepsilon_{k-i}, \qquad \tilde{s}_{k+1} = \mu + \sum_{i=1}^{q} \theta_i \, \varepsilon_{k+1-i}$$

ε_k: innovations, estimated as $s_k - \tilde{s}_k$; θ_i: MA coefficients; μ: mean; q: order



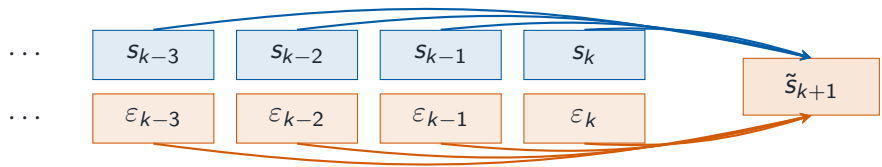
- a shock ε_k affects the next q forecasts only, then the forecast is μ
- innovations unobserved: fitted by maximum likelihood

G. E. P. Box, G. M. Jenkins, G. C. Reinsel, G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed., Wiley (2015)

ARMA(p, q) Model

ARMA(p, q): p past values (blue), q past innovations (orange)

$$\tilde{s}_{k+1} = \phi_0 + \sum_{i=1}^{p} \phi_i \, s_{k+1-i} + \sum_{i=1}^{q} \theta_i \, \varepsilon_{k+1-i}$$



- few parameters where AR needs many; orders by the Akaike information criterion (AIC); MA terms describe correlated noise, these records are nearly deterministic

P. J. Brockwell, R. A. Davis, *Introduction to Time Series and Forecasting*, Springer (2016), doi

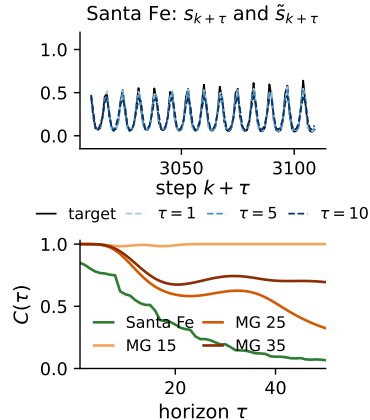
AR Forecast τ Steps Ahead

Direct AR

$$\tilde{s}_{k+\tau} = \mathbf{w}_\tau^\top \mathbf{u}_k + b_\tau$$

$\mathbf{u}_k$: window of the last L values; $\mathbf{w}_\tau \in \mathbb{R}^L$, b_τ: weights and offset, one set per horizon τ, by ridge regression on the training block; L on the validation block

- Mackey–Glass: $C(\tau)$ stays high over many steps
- Santa Fe: C below 0.5 after 16 steps, the collapses are missed



top: Santa Fe target $s_{k+\tau}$ and forecasts at $\tau = 1, 5, 10$ against $k+\tau$; bottom: $C(\tau)$ on the test block

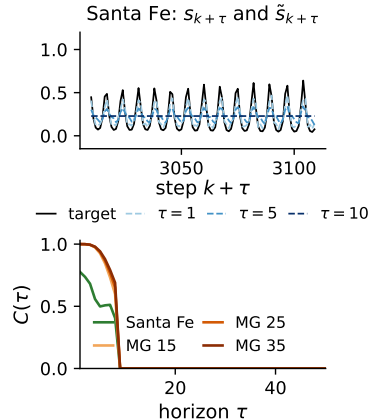
MA Forecast τ Steps Ahead

MA(q)

$$\tilde{s}_{k+1} = \mu + \sum_{i=1}^{q} \theta_i \, \varepsilon_{k+1-i}, \qquad \varepsilon_k = s_k - \tilde{s}_k$$

μ: mean of the signal, fitted with θ_i on the training block; ε_k: estimated innovations; τ steps: future innovations set to 0; here $q = 8$ (AIC)

- beyond q steps the forecast is the mean μ: no memory of the values
- weakest model on the four signals shown



top: Santa Fe target $s_{k+\tau}$ and forecasts at $\tau = 1, 5, 10$ against $k + \tau$; bottom: $C(\tau)$ on the test block

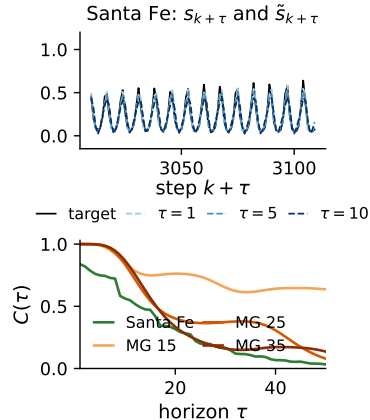
ARMA Forecast τ Steps Ahead

ARMA(p, q)

$$\tilde{s}_{k+1} = \phi_0 + \sum_{i=1}^{p} \phi_i \, s_{k+1-i} + \sum_{i=1}^{q} \theta_i \, \varepsilon_{k+1-i}$$

ϕ_0: constant; ϕ_i, θ_i: fitted on the training block; τ steps: the one-step model iterated, future innovations set to 0

- close to AR: on these deterministic signals the MA part adds little
- iterated: errors of each step feed the next



top: Santa Fe target $s_{k+\tau}$ and forecasts at $\tau = 1, 5, 10$ against $k + \tau$; bottom: $C(\tau)$ on the test block

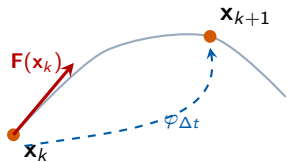
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Making Nonlinear Dynamics Linear



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From the Velocity F to the Flow Map $\varphi_{\Delta t}$



red: velocity now, the tangent; blue: where the state is after Δt

$\mathbf{F}$: velocity now

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$$

$\varphi_{\Delta t}$: position after Δt

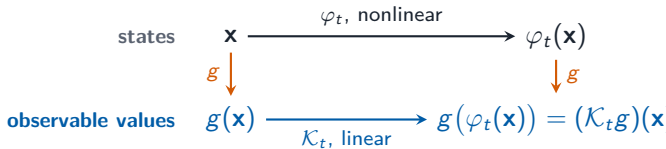
$$\varphi_{\Delta t}(\mathbf{x}) = \mathbf{x} + \int_0^{\Delta t} \mathbf{F}(\mathbf{x}(s))\,\mathrm{d}s$$

nonlinear for a nonlinear $\mathbf{F}$: no matrix advances the state

Properties of the flow map

$$\varphi_0 = \mathrm{id}, \qquad \varphi_{t+s} = \varphi_t \circ \varphi_s, \qquad \mathbf{x}_{k+n} = \varphi_{n\Delta t}(\mathbf{x}_k)$$

Koopman Operator



Definition

observable $g : \mathbb{R}^d \to \mathbb{C}$

x_1, energy, $x_1 x_3$; the measurement h

$$(\mathcal{K}_t g)(\mathbf{x}) \coloneqq g(\varphi_t(\mathbf{x}))$$

B. O. Koopman, PNAS 17, 315 (1931)

Linear for any **F**

$$\mathcal{K}_t(c_1 g_1 + c_2 g_2) = c_1 \mathcal{K}_t g_1 + c_2 \mathcal{K}_t g_2$$

$$\mathcal{K}_{t+s} = \mathcal{K}_t \mathcal{K}_s, \qquad \mathcal{K} \coloneqq \mathcal{K}_{\Delta t}$$

Generator and Eigenfunctions

Generator

$$\mathcal{A}g := \mathbf{F} \cdot \nabla g, \qquad \mathcal{K}_t = e^{t\mathcal{A}}$$

rate of change of g along the flow

Eigenfunction ξ, eigenvalue Λ

$$\mathcal{A}\xi = \Lambda\xi \;\Leftrightarrow\; \xi(\mathbf{x}(t)) = e^{\Lambda t}\,\xi(\mathbf{x}(0))$$

sampled: $\nu := e^{\Lambda \Delta t}$, $\Lambda = \ln \nu / \Delta t$

Harmonic oscillator

$\dot{x} = v,\ \dot{v} = -\omega_0^2 x$

$\xi_\pm = x \pm \mathrm{i}v/\omega_0$: $\Lambda = \mp \mathrm{i}\omega_0$

energy $= \frac{1}{2}\omega_0^2\,\xi_+\xi_-$: $\Lambda = 0$

- $\mathrm{Re}\,\Lambda$: growth rate; $\mathrm{Im}\,\Lambda$: frequency
- products of eigenfunctions: summed Λ
- $g = \sum_j c_j \xi_j \Rightarrow$ exact linear forecast

Nonlinear Map with a State-Dependent Matrix

Nonlinear map in (x, y)

$$x_{k+1} = a\,x_k, \qquad y_{k+1} = b\,y_k + c\,x_k^2$$

a, b, c: real parameters of this example only

Written with a matrix

$$\begin{pmatrix} x \\ y \end{pmatrix}_{k+1} = \begin{pmatrix} a & 0 \\ c\,x_k & b \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}_k$$

the entry $c\,x_k$ depends on the state

- no constant 2×2 matrix advances (x, y)
- the term x_k^2 is the obstacle

Lifting to Three Observables

Add x^2 as a third coordinate

$\mathbf{q} \coloneqq (x,\ y,\ x^2)$

$$x_{k+1} = a\,x_k$$
$$y_{k+1} = b\,y_k + c\,x_k^2$$
$$x_{k+1}^2 = a^2\,x_k^2$$

every right side is a combination of x, y, x^2: invariant span

A constant 3×3 matrix

$$\mathbf{q}_{k+1} = \underbrace{\begin{pmatrix} a & 0 & 0 \\ 0 & b & c \\ 0 & 0 & a^2 \end{pmatrix}}_{K} \mathbf{q}_k$$

matrix power and direct iteration agree to 4×10^{-16} for $a = 0.9,\ b = 0.5,\ c = 1$

Eigenfunctions from left eigenvectors

$$\xi_a = x, \quad \xi_{a^2} = x^2, \quad \xi_b = y + \frac{c}{b - a^2}\,x^2$$

Larger Closed Dictionary

y^2 needs x^2y and x^4

$$y^2 \mapsto b^2y^2 + 2bc\,x^2y + c^2x^4$$
$$x^2y \mapsto a^2b\,x^2y + a^2c\,x^4$$
$$x^4 \mapsto a^4x^4$$

- $(x, y, x^2, y^2, x^2y, x^4)$: closed, 6×6 matrix K_6
- eigenvalues $a, b, a^2, b^2, a^2b, a^4$: products
- $x_{k+1} = ax_k + ex_k^2$: $x^n \mapsto \ldots x^{2n}$, never closes

EDMD closure error: (x, y, x^2, y^2) leaves 0.193 in y^2; the six functions close to 1×10^{-15}

Koopman Eigenvalues of the Pendulum

Damped pendulum

$$\dot\theta = \omega, \quad \dot\omega = -\omega_p^2 \sin\theta - \gamma\omega$$

$\omega_p = 1$, $\gamma = 0.2$; hanging position: stable spiral

Principal eigenvalues = Jacobian eigenvalues

$\Lambda_\pm = -\frac{\gamma}{2} \pm \mathrm{i}\sqrt{\omega_p^2 - \gamma^2/4} = -0.1 \pm 0.99499\,\mathrm{i}$

eigenfunctions $\approx$ linear near the fixed point, defect $\omega_p^2(\theta - \sin\theta)$

A. Mauroy, I. Mezić, J. Moehlis, Physica D 261, 19 (2013)

Energy $E = \frac{1}{2}\omega^2 + \omega_p^2(1 - \cos\theta)$

$$\mathcal{A}E = -\gamma\,\omega^2$$

$\gamma = 0$: E is an eigenfunction, $\Lambda = 0$

- friction: one pair $\Lambda_\pm$, lattice $n_+\Lambda_+ + n_-\Lambda_-$
- no friction: frequency $\Omega(E)$ of each swing, a band
- limit cycle: $\Lambda = \mathrm{i}n\Omega$; chaos: continuous spectrum

Finite Dictionary Fitted to Data

EDMD: least squares with ridge β

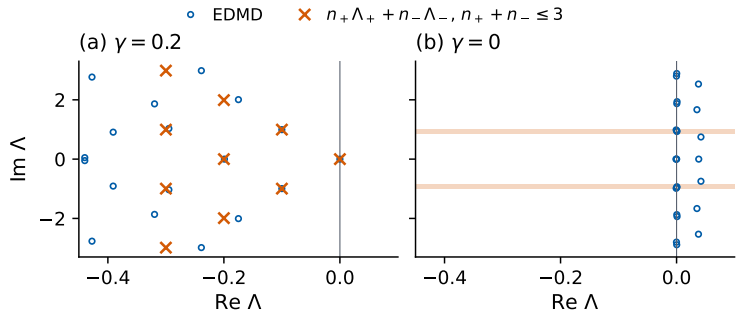
$$K = \Sigma_1(\Sigma_0 + \beta I)^{-1}, \qquad \Sigma_0 = \frac{1}{n_p}\sum_i \mathbf{g}(\mathbf{x}_i)\mathbf{g}(\mathbf{x}_i)^\top, \quad \Sigma_1 = \frac{1}{n_p}\sum_i \mathbf{g}(\mathbf{x}'_i)\mathbf{g}(\mathbf{x}_i)^\top$$

$\mathbf{x}'_i$: state one step after $\mathbf{x}_i$; DMD: $\mathbf{g}(\mathbf{x}) = \mathbf{x}$

- closure error $\mathbf{g}(\mathbf{x}') - K\mathbf{g}(\mathbf{x})$: zero only for an invariant span
- not closed: a projection, eigenvalues depend on the data
- DMD on (x, y): $b = 0.5$ comes out as 0.471

P. J. Schmid, J. Fluid Mech. 656, 5 (2010); M. O. Williams *et al.*, J. Nonlinear Sci. 25, 1307 (2015)

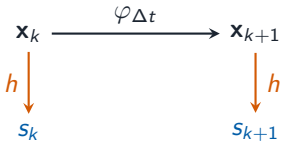
EDMD Eigenvalues of the Pendulum



$\Lambda = \ln \nu / \Delta t$, 36 monomials in θ, ω of total degree ≤ 7, $\Delta t = 0.1$; crosses: $n_+\Lambda_+ + n_-\Lambda_-$; band: $\Omega(E)$ of the sampled swings

- $\gamma = 0.2$: $\Lambda = -0.1001 \pm 0.9942\,\mathrm{i}$
- $\gamma = 0$: energy found, frequencies scattered, spurious growth

Koopman and the Measured Series



hidden state above, measured values below

The measurement is an observable

$$(\mathcal{K}h)(\mathbf{x}_k) = h(\mathbf{x}_{k+1}) = s_{k+1}$$
$$s_{k+n} = (\mathcal{K}^n h)(\mathbf{x}_k)$$

$\mathcal{K}$ acts on the function h; s_k is one of its values

- window $\mathbf{u}_k = (s_k, \ldots, s_{k-L+1})$: a dictionary generated by h
- AR: past values; NVAR: plus products; networks: learned observables

Linear Map Fitted to the Delay Window

Delay DMD

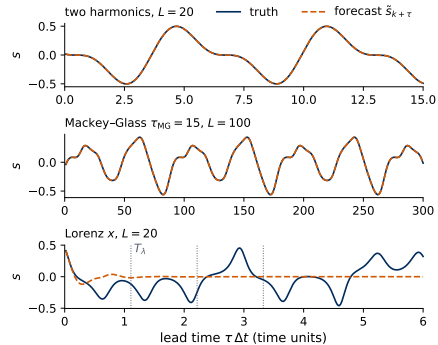
$$K = \arg\min_K \sum_k \left\| \mathbf{u}_{k+1} - K\mathbf{u}_k \right\|^2$$

window $L = 20$ (MG 15: 100)

Companion matrix

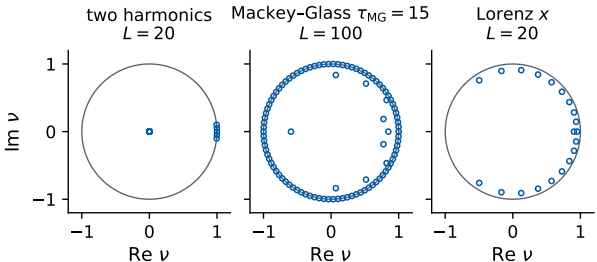
$$K = \begin{pmatrix} \phi_1 & \cdots & \phi_L \\ 1 & & 0 \\ & \ddots & \end{pmatrix}$$

first row: AR(L); eigenvalues $\nu_j = 1/z_j$, z_j: roots of $1 - \sum_i \phi_i z^i$



dashed: K^n from one window; top: $\sin t + \frac{1}{2}\sin 2t$; middle: MG 15; bottom: Lorenz, dotted: multiples of T_λ

Eigenvalues of the Window Maps



eigenvalues ν and the unit circle; $L = 20, 100, 20$

- two harmonics: ν on the circle at $0, \pm 1, \pm 2$
- MG 15: many harmonics, $|\nu| \leq 1.00005$
- Lorenz: all inside, $|\nu| \leq 0.951$, decay

Limits

closure error; conditioning (Lorenz, $L = 30$: condition number 3×10^9, $|\nu|_{\max} = 1.0107$ or 0.9635 by solver); noise in the inputs

4

Nonlinear Vector Autoregression



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Nonlinear Features of the Window

A linear readout is a flat plane through the data; products of past values let it bend.

Nonlinear vector autoregression (NVAR)

$$\varphi_k = \big(1,\ \underbrace{s_k, s_{k-\kappa}, \ldots, s_{k-(L-1)\kappa}}_{\text{linear}},\ \underbrace{s_{k-i\kappa}\, s_{k-j\kappa}\ (0 \le i \le j \le L-1)}_{\text{quadratic}}\big), \qquad \tilde{s}_{k+\tau} = \mathbf{w}_\tau^\top \varphi_k$$

- **trainable parameters**: the readout weights $\mathbf{w}_\tau$, one vector for every horizon τ
- the features are fixed, so training is again one least-squares problem
- L values spaced by κ samples, both chosen on the validation block

D. J. Gauthier, E. Bollt, A. Griffith, W. A. S. Barbosa, Nat. Commun. 12, 5564 (2021), doi:10.1038/s41467-021-25801-2

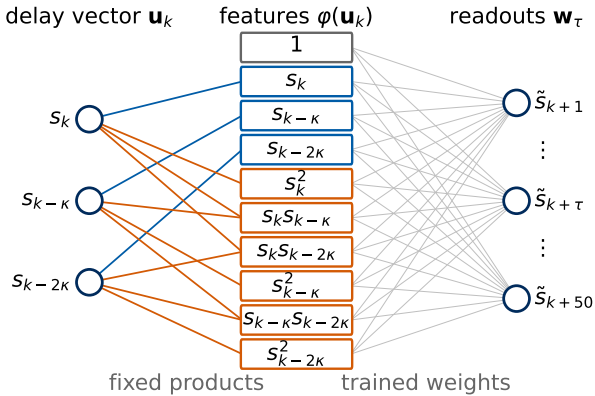
NVAR with Two Past Values

For $L = 2$ the forecast is written out in full,

$$\tilde{s}_{k+1} = w_0 + \underbrace{w_1 s_k + w_2 s_{k-1}}_{\text{as in AR(2)}} + \underbrace{w_3 s_k^2 + w_4 s_k s_{k-1} + w_5 s_{k-1}^2}_{\text{new: products}}$$

- with the products the forecast can curve: the effect of s_k depends on the values themselves, so it can follow a curved cloud instead of a flat plane
- a window of L values gives $1 + L + L(L+1)/2$ features, 6 for $L = 2$ and 153 for $L = 16$
- the number of features grows quickly with L, so the weights need a penalty

NVAR Feature Map



Feature map for $L = 3$: each input feeds its linear feature (blue) and the products that contain it (orange).

- only the weights to the readouts are trained
- the number of features grows as L^2, 153 at $L = 16$

Ridge Readout

With hundreds of features, plain least squares fits the noise.

Ridge regression

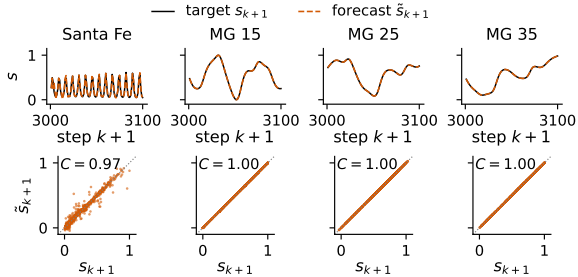
$$\mathbf{w}_\tau = \arg\min_{\mathbf{w}} \sum_k \left(s_{k+\tau} - \mathbf{w}^\top \boldsymbol{\varphi}_k\right)^2 + \beta \|\mathbf{w}\|^2 = \left(X^\top X + \beta I\right)^{-1} X^\top \mathbf{y}$$

$\beta > 0$: ridge strength; X: features $\boldsymbol{\varphi}_k$ as rows; $\mathbf{y}$: targets $s_{k+\tau}$

- the penalty keeps the weights small when features are nearly redundant
- window, spacing κ and β are chosen together on the validation block

A. E. Hoerl, R. W. Kennard, Technometrics 12, 55 (1970), doi:10.1080/00401706.1970.10488634

One-Step NVAR Forecasts

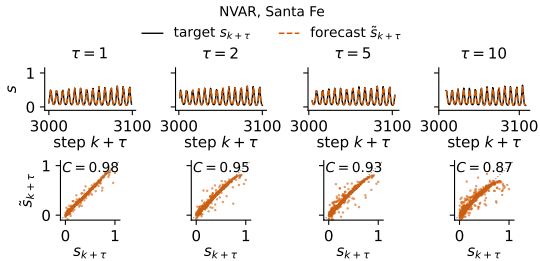


top: target s_{k+1} (solid) and forecast $\tilde{s}_{k+1}$ (dashed) against the step $k+1$, first 100 test targets; bottom: $\tilde{s}_{k+1}$ against s_{k+1} for the whole test block, $C = r^2$ in each panel; L, κ, β chosen on the validation block (Santa Fe: $L = 32$, $\kappa = 2$, $\beta = 10^{-2}$), readout refitted on the training and validation blocks

- Santa Fe: the points stay near the diagonal also at the high flashes, where the linear models bend away

NVAR Forecast τ Steps Ahead

$$\tilde{s}_{k+\tau} = \mathbf{w}_\tau^\top \boldsymbol{\varphi}_k \quad \text{one ridge weight vector per horizon } \tau$$



top: target $s_{k+\tau}$ (solid), forecast $\tilde{s}_{k+\tau}$ (dashed) against the step $k + \tau$; bottom: $\tilde{s}_{k+\tau}$ against $s_{k+\tau}$, test block, with C

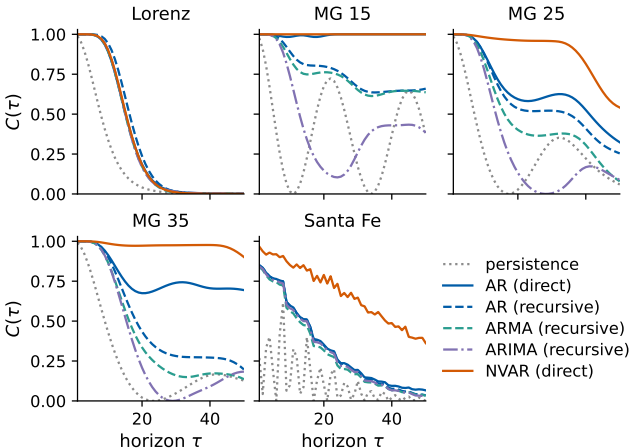
- $C = 0.98$ at $\tau = 1$, 0.87 at $\tau = 10$; direct AR 0.85 and 0.60
- fit as for the networks: $L = 32$, $\kappa = 1$, training block only; one-step fit with $\kappa = 2$ and validation refit: $C = 0.97$ at $\tau = 1$

5
Comparisons



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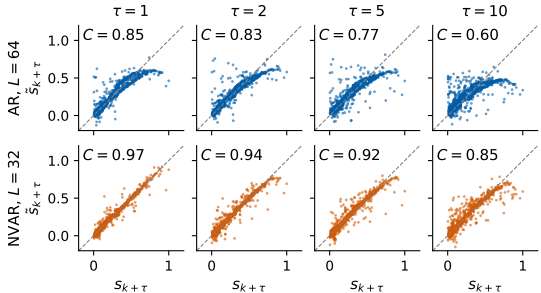
Capacity of the Linear Models and NVAR



$C(\tau)$ on the test block; dotted: persistence $\tilde{s}_{k+\tau} = s_k$; solid: direct AR (blue), NVAR (orange); dashed: recursive AR, ARMA, ARIMA

- MG 15: direct AR and NVAR near $C = 1$ at all horizons
- chaotic MG: NVAR far beyond the linear models
- Lorenz: AR and NVAR below $C = 0.5$ after 16 samples, well inside $T_\lambda = 55$ samples

Limits of Linear Models



Santa Fe test block: forecast $\tilde{s}_{k+\tau}$ against target $s_{k+\tau}$ at $\tau = 1, 2, 5, 10$; top: direct AR, $L = 64$; bottom: NVAR, $L = 32$; dotted: diagonal; C in each panel

- AR underestimates the large flashes and its cloud widens with τ; NVAR stays near the diagonal
- NVAR features are fixed in advance, grow fast with L, and remember only a fixed window

6

Neural Networks

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Neural Networks for Forecasting

Every forecast so far is a linear readout of features of the past, $\tilde{s}_{k+\tau} = \mathbf{w}_\tau^\top \Phi(\text{past})$. Networks learn Φ.

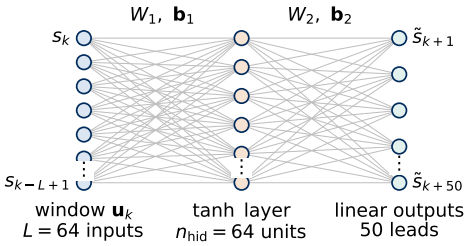
1. **multilayer perceptron (MLP)**: NVAR with learned features; memory: the window of the last L values

2. **recurrent neural network (RNN)**: a learned dynamical system, a state $\mathbf{h}_k$ updated from $\mathbf{h}_{k-1}$ and s_k; gradients through the unrolled steps vanish or explode

3. **long short-term memory (LSTM), gated recurrent unit (GRU)**: gates add a memory path that can keep values and gradients over many steps; trained on the laser, the LSTM forget gates keep a median memory time of only 2.5 steps

Santa Fe	AR	NVAR	MLP	RNN	GRU	LSTM
C_{tot} of 50	16.5	34.7	35.3	42.5	43.0	44.3

MLP Forecaster



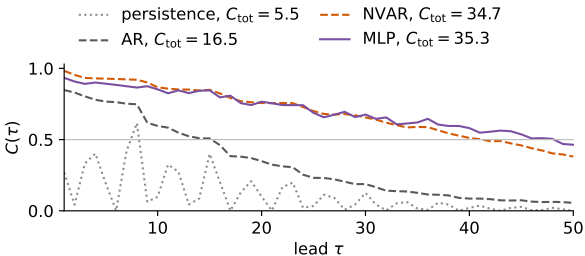
Multilayer perceptron (MLP)

$$\tilde{\mathbf{y}}_k = W_2 \tanh\left(W_1 \mathbf{u}_k + \mathbf{b}_1\right) + \mathbf{b}_2$$

$\mathbf{u}_k$: window of the last L values;
$\tilde{\mathbf{y}}_k = (\tilde{s}_{k+1}, \ldots, \tilde{s}_{k+50})$: one output per horizon

- NVAR with learned features $\tanh(W_1 \mathbf{u}_k + \mathbf{b}_1)$
- $L = n_{\text{hid}} = 64$: 7410 parameters, gradient descent on all 50 horizons
- memory limited to the window

MLP Forecast Capacity

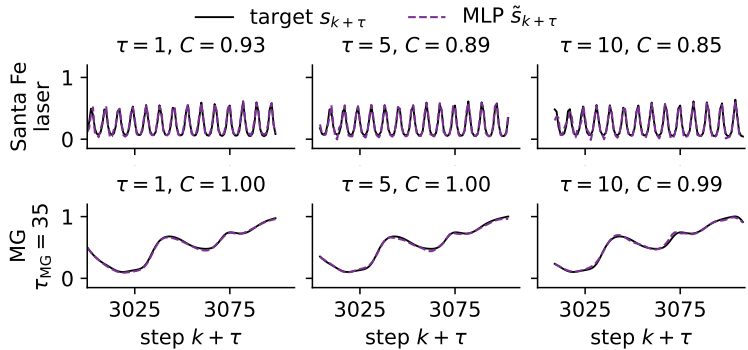


C_{tot}	AR	NVAR	MLP
Santa Fe	16.5	34.7	**35.3**
MG 15	49.8	50.0	**50.0**
MG 25	33.1	45.5	**48.4**
MG 35	38.0	48.6	**49.5**

Left: $C(\tau)$ on the Santa Fe test block; dotted, **persistence** $\tilde{s}_{k+\tau} = s_k$: repeat the last value, good only when τ is near a multiple of the pulse period (7.26 samples). Table: $C_{\text{tot}} = \sum_{\tau=1}^{50} C(\tau)$.

- NVAR better at $\tau = 1$ (0.98 vs 0.93), MLP at $\tau = 50$ (0.46 vs 0.38)
- Santa Fe: MLP lead over NVAR about its seed spread, $C_{\text{tot}} = 35.3$–36.2 (5 seeds)

MLP Forecasts on the Laser and Mackey–Glass



black: target $s_{k+\tau}$; dashed: MLP forecast $\tilde{s}_{k+\tau}$, first 100 test steps, horizons $\tau = 1, 5, 10$; C on the whole test block; MG 15, 25 (not shown): $C \geq 0.99$ up to $\tau = 10$

- Santa Fe: C drops from 0.93 at $\tau = 1$ to 0.85 at $\tau = 10$, the pulse heights are missed first; MG 35: $C \geq 0.99$

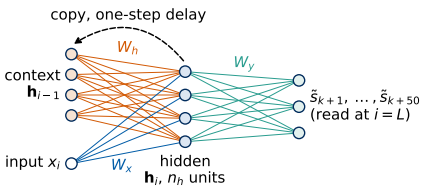
7

Recurrent Neural Networks



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Recurrent Cell



Recurrent neural network (RNN)

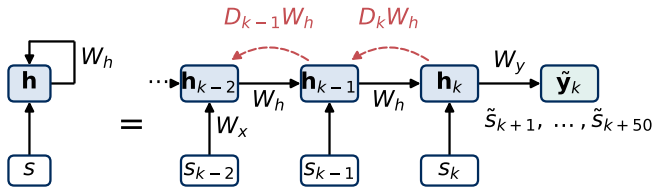
$$\mathbf{h}_k = \tanh\left(W_h \mathbf{h}_{k-1} + W_x s_k + \mathbf{b}\right), \quad \tilde{\mathbf{y}}_k = W_y \mathbf{h}_k + \mathbf{b}_y$$

$W_h \in \mathbb{R}^{n_h \times n_h}$: feedback; $W_x \in \mathbb{R}^{n_h}$: input; $W_y, \mathbf{b}_y$: readout of the 50 horizons

- the state $\mathbf{h}_k$ is fed back at the next step, so it depends on $s_k, s_{k-1}, \ldots$
- the parameter count does not grow with the history; $\mathbf{h}_k \in (-1, 1)^{n_h}$

J. L. Elman, Cogn. Sci. 14, 179 (1990), doi:10.1207/s15516709cog1402_1

Forecasting with a Recurrent Network



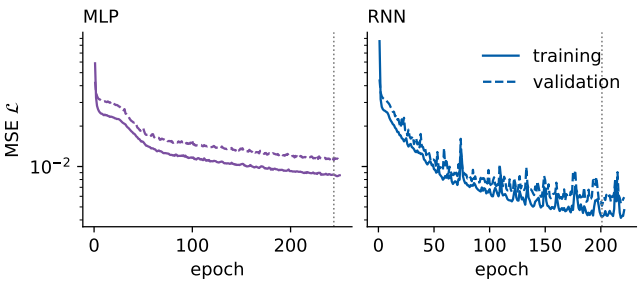
Update and readout

$$\mathbf{h}_k = \tanh\left(W_h\mathbf{h}_{k-1} + W_x s_k + \mathbf{b}\right), \qquad \tilde{\mathbf{y}}_k = W_y\mathbf{h}_k + \mathbf{b}_y = (\tilde{s}_{k+1}, \ldots, \tilde{s}_{k+50})$$

left: cell with feedback W_h; right: unrolled, one copy per input, all sharing W_h, W_x, $\mathbf{b}$; red: backward dependencies $D_i W_h$, $D_i = \mathrm{diag}(1 - \mathbf{h}_i^2)$

- $\mathbf{h} = 0$ at each origin, feed the window of $L = 64$ values, read all 50 horizons from the last state
- unrolled: a deep network, one layer per input value

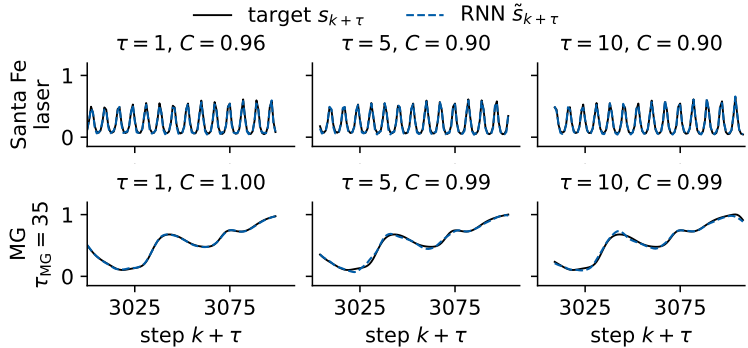
Training Curves on the Laser



Mean-square error $\mathcal{L}$ over the 50 horizons against the training epoch on Santa Fe, solid for training and dashed for validation. The dotted line marks the epoch with the lowest validation loss, whose weights are kept.

- the RNN reaches a lower loss than the MLP on the same windows

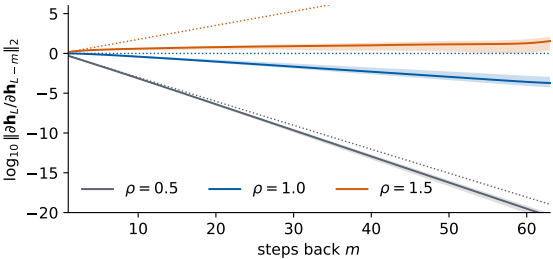
RNN Forecasts on the Laser and Mackey–Glass



black: target $s_{k+\tau}$; dashed: RNN forecast $\tilde{s}_{k+\tau}$, first 100 test steps, horizons $\tau = 1, 5, 10$; C on the whole test block; MG 15, 25 (not shown): $C \geq 0.99$ up to $\tau = 10$

- Santa Fe: C from 0.96 to 0.90, above the MLP at every horizon shown; MG 35: $C \geq 0.99$

Unstable Gradients and Remedies



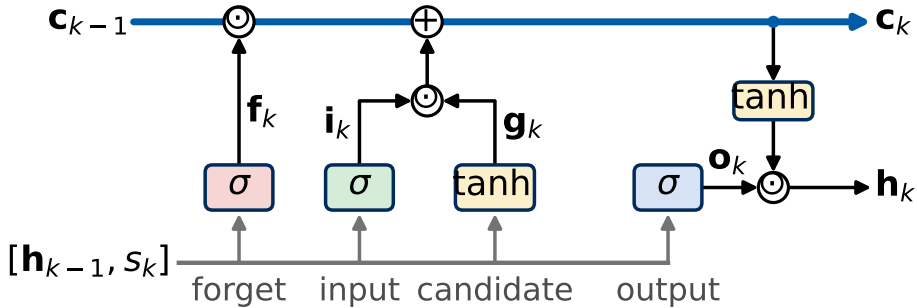
$\log_{10} \|\partial \mathbf{h}_k / \partial \mathbf{h}_{k-n}\|_2$ against the steps back n, written $\partial \mathbf{h}_L / \partial \mathbf{h}_{L-m}$ and m in the drawing, untrained RNNs, 10 per value ϱ (ρ in the legend) of all singular values of W_h; dotted: slope $\log_{10} \varrho$ per step

- $\partial \mathbf{h}_k / \partial \mathbf{h}_{k-n} = \prod_i D_i W_h$, typically $\sim e^{\lambda_h n}$
- $\lambda_h < 0$: vanishing; $\lambda_h > 0$: exploding
- $\lambda_h = -0.755$, -0.146, $+0.030$ per step at $\varrho = 0.5, 1, 1.5$

Remedies: clipping (exploding), orthogonal W_h, gated cells (vanishing)

Bengio *et al.*, IEEE TNN 5, 157 (1994); Pascanu *et al.*, arXiv:1211.5063; Saxe *et al.*, arXiv:1312.6120

LSTM Cell



top line: cell state $\mathbf{c}_{k-1} \to \mathbf{c}_k$, changed only by the forget gate $\mathbf{f}_k$ and the gated candidate $\mathbf{i}_k \odot \mathbf{g}_k$; $\mathbf{h}_k$: cell state through tanh and the output gate $\mathbf{o}_k$; inputs s_k, $\mathbf{h}_{k-1}$

- second state $\mathbf{c}_k$, changed only by multiplication and addition
- three gates with values in $(0, 1)$ decide what is kept, written and shown

S. Hochreiter, J. Schmidhuber, Neural Comput. 9, 1735 (1997)

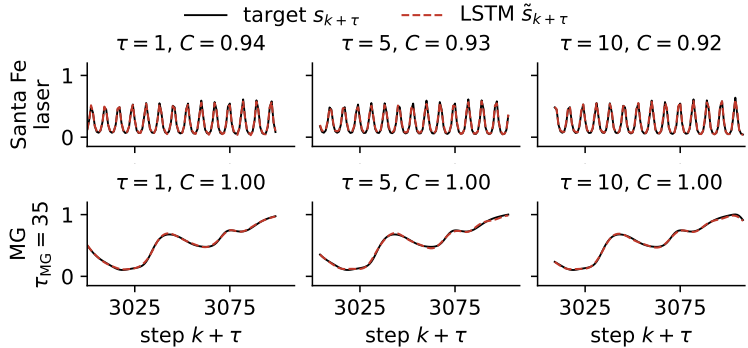
LSTM Equations

$$
\begin{aligned}
\mathbf{f}_k &= \sigma(W_f[\mathbf{h}_{k-1}, s_k] + \mathbf{b}_f) && \text{forget gate} \\
\mathbf{i}_k &= \sigma(W_i[\mathbf{h}_{k-1}, s_k] + \mathbf{b}_i) && \text{input gate} \\
\mathbf{g}_k &= \tanh(W_g[\mathbf{h}_{k-1}, s_k] + \mathbf{b}_g) && \text{candidate memory} \\
\mathbf{c}_k &= \mathbf{f}_k \odot \mathbf{c}_{k-1} + \mathbf{i}_k \odot \mathbf{g}_k && \text{cell-memory update} \\
\mathbf{o}_k &= \sigma(W_o[\mathbf{h}_{k-1}, s_k] + \mathbf{b}_o) && \text{output gate} \\
\mathbf{h}_k &= \mathbf{o}_k \odot \tanh(\mathbf{c}_k) && \text{hidden-state output}
\end{aligned}
$$

$[\mathbf{h}_{k-1}, s_k]$: old state and new value stacked, $n_h + 1$ numbers; $\sigma(x) = 1/(1 + e^{-x})$; $\odot$: product component by component

- direct path $\partial \mathbf{c}_k / \partial \mathbf{c}_{k-1} = \mathrm{diag}(\mathbf{f}_k)$: no weight matrix; with $\mathbf{f}_k \approx f$ a value decays as f^n, memory time $\tau_{\mathrm{mem}} = -1/\ln f$
- trained on the laser: median $\tau_{\mathrm{mem}} = 2.5$ steps, largest 3.5; it counts only the direct retention along $\mathbf{c}$, information also travels through $\mathbf{h}$ and the gates

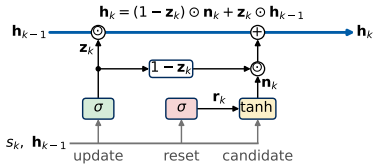
LSTM Forecasts on the Laser and Mackey–Glass



black: target $s_{k+\tau}$; dashed: LSTM forecast $\tilde{s}_{k+\tau}$, first 100 test steps, horizons $\tau = 1, 5, 10$; C on the whole test block; MG 15, 25 (not shown): $C \geq 0.99$ up to $\tau = 10$

- Santa Fe: C from 0.94 to 0.92, the flattest decay of the three networks; MG 35: $C \geq 0.99$

GRU Cell



top line: state $\mathbf{h}_{k-1} \to \mathbf{h}_k$; green: update gate, red: reset gate, yellow: candidate

$$\mathbf{z}_k = \sigma(\mathbf{w}_z s_k + W_{hz}\mathbf{h}_{k-1} + \mathbf{b}_z) \quad \text{update gate}$$

$$\mathbf{r}_k = \sigma(\mathbf{w}_r s_k + W_{hr}\mathbf{h}_{k-1} + \mathbf{b}_r) \quad \text{reset gate}$$

$$\mathbf{n}_k = \tanh\left(\mathbf{w}_n s_k + \mathbf{b}_n + \mathbf{r}_k \odot (W_{hn}\mathbf{h}_{k-1})\right) \quad \text{candidate state}$$

$$\mathbf{h}_k = (1 - \mathbf{z}_k) \odot \mathbf{n}_k + \mathbf{z}_k \odot \mathbf{h}_{k-1} \quad \text{hidden-state update}$$

- $\mathbf{z}_k \approx 1$: the unit keeps its old state; reset applied after the matrix, as in common libraries
- 4914 parameters (LSTM 6002, $n_h = 32$); Santa Fe $C_{\text{tot}} = 43.0$, between RNN 42.5 and LSTM 44.3, seeds 43.0–43.5

K. Cho *et al.*, EMNLP (2014), doi:10.3115/v1/D14-1179

All Models on All Signals

C_{tot}	Lorenz x	MG 15	MG 25	MG 35	Santa Fe
persistence	8.4	19.5	13.3	11.8	5.5
AR	15.4	49.8	33.1	38.0	16.5
NVAR	15.4	50.0	45.5	48.6	34.7
MLP	29.4	50.0	48.4	49.5	35.3
RNN	43.8	50.0	49.4	49.7	42.5
GRU	45.1	50.0	49.8	49.9	43.0
LSTM	43.8	50.0	49.9	49.9	44.3

$C_{\mathrm{tot}} = \sum_{\tau=1}^{50} C(\tau) \leq 50$; all models fitted on the same origins and scored on the same 951 test origins (AR, NVAR differ slightly from the values above)

- MG 15: all near 50; laser, chaotic MG: AR lowest, NVAR and MLP higher, recurrent networks highest (MG 35: MLP and recurrent networks all within 0.5 of 50)
- Lorenz: AR, NVAR below $C = 0.5$ after 16 samples, well inside $T_\lambda = 55$ samples; RNN to LSTM keep $C(50) = 0.74$–0.77

8

Neural Ordinary Differential Equations



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From the Record to the Laws of Motion

So far: the record only

models $\tilde{G}_\tau$ of the forecast map G_τ, learned from $s_1, \ldots, s_k$; they need long records and know nothing beyond the data

Often more is known

the whole state can be measured, and the law of motion is known in form, in part or completely: Newton's equation, a diffusion equation, a conservation law

- form known: neural ordinary differential equation (Neural ODE)
- whole law known: physics-informed neural network (PINN)
- one term missing: hybrid model

Learning the Equation of Motion

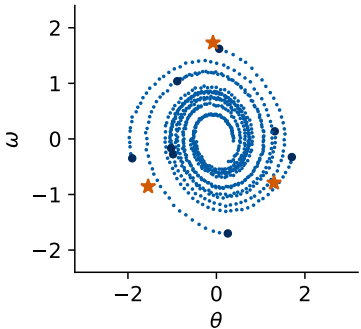
Dynamical system

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}(t) = \mathbf{F}(\mathbf{x}(t)), \qquad \mathbf{x}(t) = \varphi_t(\mathbf{x}_0)$$

$\mathbf{x}$: state; $\mathbf{F}$: vector field, the velocity of the state at every point of phase space; φ_t: flow map

- known: the form; unknown: $\mathbf{F}$ itself
- model: $\mathbf{F} \approx \mathbf{F}_\theta$, a neural network with input $\mathbf{x}$ and output $\dot{\mathbf{x}}$
- data: measured states $\mathbf{y}_k$ at times t_k; θ chosen so that the solution of $\dot{\mathbf{x}} = \mathbf{F}_\theta(\mathbf{x})$ passes through them

Learning Task on the Damped Pendulum



$(\theta, \omega = \dot{\theta})$; blue: measured states of 8 trajectories, every 0.1, noise $\sigma = 0.01$; dark circles: starts; stars: test starts

measured: states $\mathbf{y}_k$ only, no derivatives

↓

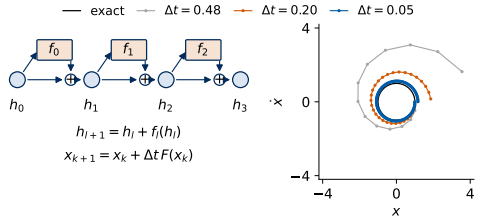
learned: field $\mathbf{F}_\theta(\theta, \omega)$, its values fixed only through integrated segments

↓

forecast: trajectory from integrating $\mathbf{F}_\theta$ from a test start

- the data constrain $\mathbf{F}_\theta$ in the band of spirals around the origin
- near the saddles $\theta = \pm\pi$ no data: there the field is an extrapolation

From Euler Steps to Residual Networks



Two updates of the same form

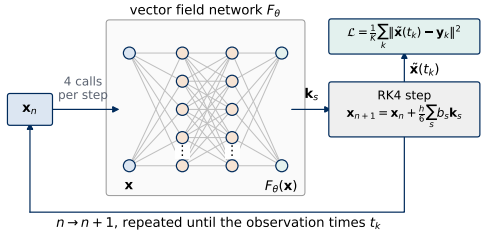
$\mathbf{h}_{l+1} = \mathbf{h}_l + \mathbf{f}_l(\mathbf{h}_l)$ (residual network), $\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t\, \mathbf{F}(\mathbf{x}_n)$ (Euler step)

right: Euler steps of $\ddot{x} = -x$ over one period in $(x, \dot{x})$, $\Delta t = 0.48, 0.20, 0.05$; each step enlarges areas by $1 + \Delta t^2$: spiral outwards

- shared layers $\mathbf{f}_l = \Delta t\, \mathbf{F}_\theta$, $\Delta t \to 0$: the layers become a flow $\dot{\mathbf{x}} = \mathbf{F}_\theta(\mathbf{x})$

K. He *et al.*, arXiv:1512.03385; R. T. Q. Chen *et al.*, arXiv:1806.07366

Neural ODE



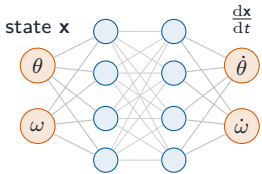
Neural ODE

$$\dot{\mathbf{x}} = \mathbf{F}_\theta(\mathbf{x}), \qquad \tilde{\mathbf{x}}(t_k) = \text{ODE solver}(\mathbf{F}_\theta, \mathbf{x}_0, t_k), \qquad \mathcal{L}(\theta) = \frac{1}{N}\sum_{k=1}^{N} \left\|\tilde{\mathbf{x}}(t_k) - \mathbf{y}_k\right\|^2$$

$\mathbf{F}_\theta$: network, input the state, output its velocity; $\mathbf{y}_k$: measured states at the N times t_k

- ODE solver here: classical fourth-order Runge–Kutta (RK4), four evaluations of $\mathbf{F}_\theta$ per step

Architecture and Training of a Neural ODE



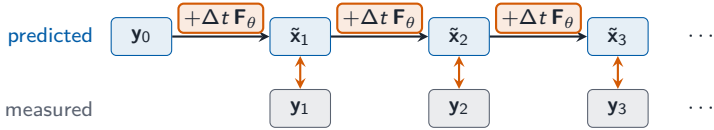
pendulum, $\mathbf{x} = (\theta, \omega)$; $2 \to 64 \to 64 \to 2$, tanh, 4482 weights

The network learns the vector field

$$\mathbf{F}_\theta(\mathbf{x}) \approx \frac{d\mathbf{x}}{dt}, \quad \text{Euler step } \mathbf{x}_{k+1} = \mathbf{x}_k + \Delta t\, \mathbf{F}_\theta(\mathbf{x}_k)$$

in the labs RK4 replaces the Euler step

result: the equation of motion $\mathbf{F}$, learned from measured trajectories; subscript θ: the weights



$$\mathcal{L}(\theta) = \sum_k \|\tilde{\mathbf{x}}_k - \mathbf{y}_k\|^2$$

$\mathbf{y}_k$: measured, $\tilde{\mathbf{x}}_k$: predicted states; same weights at every step; segments of 10 samples from measured starts, backpropagation through the solver

Gradients through the Solver

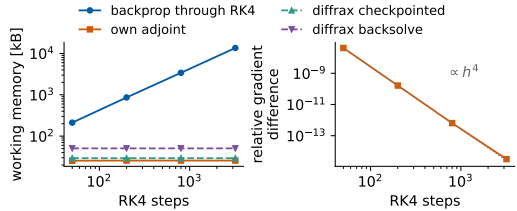
Adjoint method

$$\mathbf{a}(t) := \partial \mathcal{L} / \partial \mathbf{x}(t)$$

$$\dot{\mathbf{a}} = -J^\top \mathbf{a}, \qquad J = \partial \mathbf{F}_\theta / \partial \mathbf{x}$$

$$\frac{d\mathcal{L}}{d\theta} = \int_0^{t_N} \mathbf{a}^\top \frac{\partial \mathbf{F}_\theta}{\partial \theta}\, dt$$

$\mathbf{a}$: sensitivity of the loss to the state, integrated backwards in time from t_N, with jumps at the data times

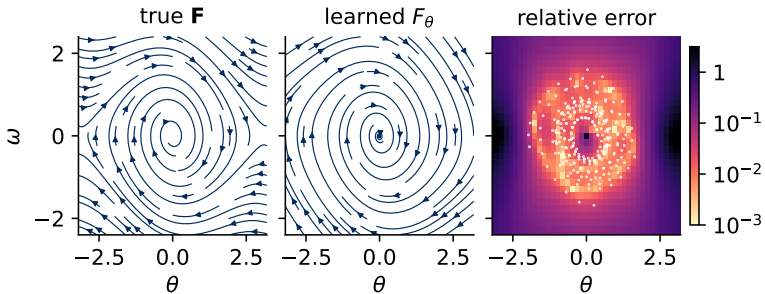


left: memory against the number of RK4 steps; right: relative difference of the two gradients; step Δt written h

- backpropagation through RK4 stores every step: 212 kB at 50 steps, 13592 kB at 3200
- adjoint: constant 26 kB; the two gradients agree to 3.2×10^{-15}, difference $\propto \Delta t^4$

P. Kidger, On Neural Differential Equations, arXiv:2202.02435

Learned Field of the Damped Pendulum

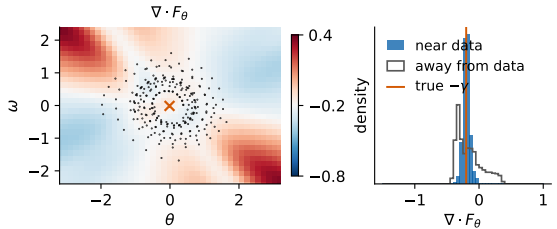


$\dot\theta = \omega$, $\dot\omega = -\omega_p^2 \sin\theta - \gamma\,\omega$, $\omega_p = 1$, $\gamma = 0.2$; left: streamlines of the true $\mathbf{F}$; middle: learned $\mathbf{F}_\theta$; right: $|\mathbf{F}_\theta - \mathbf{F}|/|\mathbf{F}|$, logarithmic colour scale, white dots: data

- median relative error 0.012 within 0.2 of the data, 0.23 farther away
- near the saddles at $\theta = \pm\pi$ the field is a guess; differentiable: its Jacobian $J_\theta = \partial \mathbf{F}_\theta / \partial \mathbf{x}$ carries physics

Divergence of the Learned Field

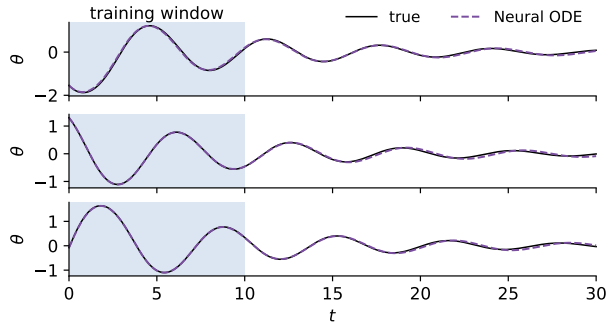
$\nabla \cdot \mathbf{F} < 0$: areas shrink, dissipative $\nabla \cdot \mathbf{F} = 0$: areas preserved



$\nabla \cdot \mathbf{F}_\theta := \operatorname{tr} J_\theta$; left: over the plane, dots: data, cross: zero of $\mathbf{F}_\theta$; right: values near and far from the data, line: true $-\gamma$; $\nabla \cdot \mathbf{F} = 0$: Hamiltonian in 2D on a region without holes, only volume preserving in more dimensions

- dissipative: $\nabla \cdot \mathbf{F}_\theta = -0.197$ near the data (quartiles $-0.229, -0.167$), true -0.2
- at $(0, 0)$: $\gamma = -\operatorname{tr} J_\theta = 0.151$, $\omega_p = \sqrt{\det J_\theta} = 0.947$ (true 0.2, 1)
- saddle at $(\pi, 0)$ not found: no data there

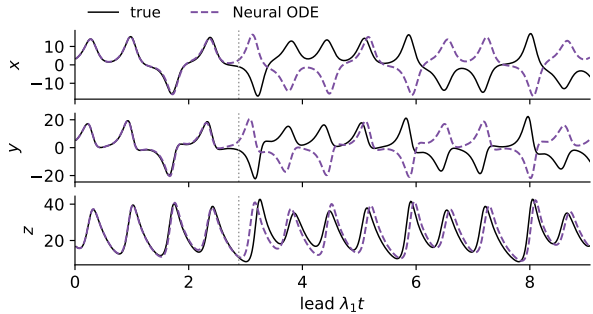
Pendulum Forecast



three test starts; trained on $0 \le t \le 10$ (shaded), integrated to $t = 30$

- largest error 8.2×10^{-2} beyond the training window, 4482 parameters

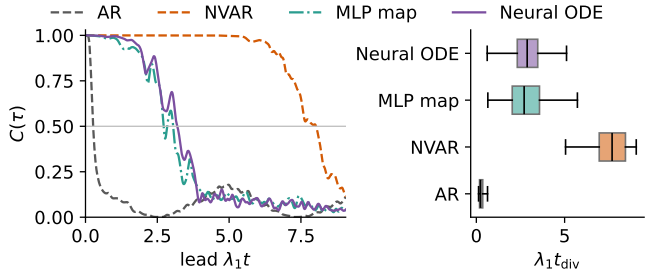
Neural ODE on Lorenz-63



x, y, z of truth and forecast against the lead in Lyapunov times $\lambda_1 t$; all three coordinates measured, $\Delta t = 0.02$, standardised; dotted: divergence time, first lead with RMS error over the coordinates above 0.5

- largest exponent: learned 0.899 ± 0.010, true field 0.906 ± 0.007
- divergence time of a few Lyapunov times, as for any model of a chaotic system

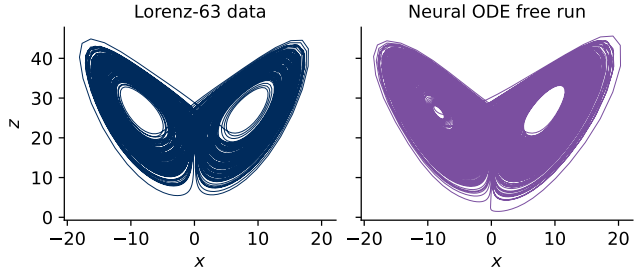
Four Models on the Lorenz State



left: $C(\tau)$ against the lead $\lambda_1 \tau \Delta t$, written $\lambda_1 t$ in the drawing, grey: $C = 0.5$; right: divergence times $\lambda_1 t_{\mathrm{div}}$ over the test origins, box: middle half, black line: median; MLP map: one state to the next; AR and NVAR on the full state

- median $\lambda_1 t_{\mathrm{div}}$: Neural ODE 2.88, MLP map 2.71, AR 0.24, NVAR 7.69
- NVAR wins: the Lorenz field is quadratic, so the one-step map of the full state is quadratic up to $O(\Delta t^2)$

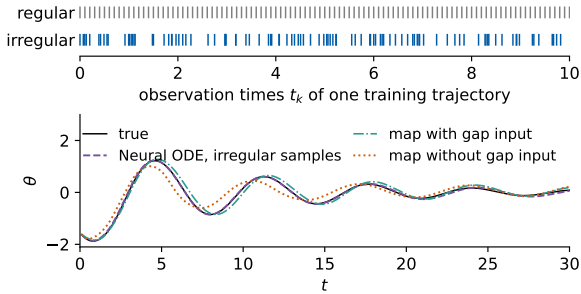
Learned Lorenz Attractor



plane (x, z); left: data; right: free run of the learned field over 400 time units

- a continuous field can be analysed: fixed points by Newton's method, Jacobians, Lyapunov exponents
- mean $\nabla \cdot \mathbf{F}_\theta$ over the test states -13.88, true -13.67
- not exact at the fixed points the data rarely visit: one convection state becomes a stable spiral, the true one is weakly unstable

Irregular Sampling



damped pendulum observed at random times; a discrete map needs a fixed step, or the gap $t_{k+1} - t_k$ as an extra input

- RMS error of θ: Neural ODE 0.033, map with gap input 0.115, map without 0.193
- the solver integrates to any requested time, no fixed step needed

9

Physics-Informed Neural Networks



JAGIELLONIAN UNIVERSITY
IN KRAKOW

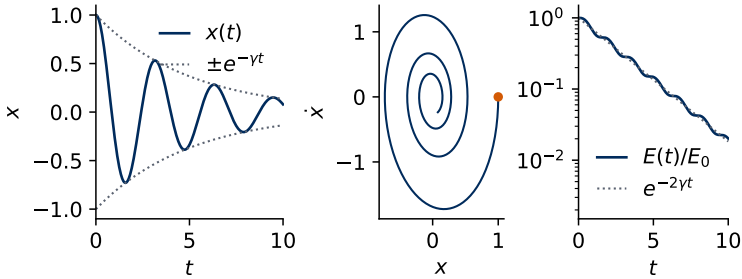
Known Laws, Few Measurements

system	known law	measured	wanted
oscillator	$m_0\ddot{x} + c\dot{x} + k_s x = 0$	few noisy positions	$x(t)$, damping c
heat in a rod	$\partial_t u = D\,\partial_x^2 u$	few thermometers	$u(x,t)$, diffusivity D
quantum particle	Schrödinger equation	few snapshots	$\psi(x,t)$, potential

- forward problem: solve the equation, here a classical solver is hard to beat
- inverse problem: find an unknown parameter of the law from data
- data assimilation: fill the gaps between sparse measurements with the law

M. Raissi, A. Yazdani, G. E. Karniadakis, Science 367, 1026 (2020); G. E. Karniadakis *et al.*, Nat. Rev. Phys. 3, 422 (2021)

Damped Harmonic Oscillator



Equation of motion

$$m_0\ddot{x} + c\dot{x} + k_s x = 0, \qquad x(0) = x_0,\ \dot{x}(0) = v_0$$

$m_0 = 1$, damping $c = 0.4$, spring constant $k_s = 4$, $x_0 = 1$, $v_0 = 0$; $\omega_0 = \sqrt{k_s/m_0}$, damping ratio $\zeta = c/(2\sqrt{k_s m_0}) = 0.10$; left: $x(t)$ in the envelope $\pm e^{-ct/(2m_0)}$, written $e^{-\gamma t}$; middle: phase portrait; right: E/E_0, $E = \frac{1}{2}m_0\dot{x}^2 + \frac{1}{2}k_s x^2$

Fitting the Data Only

Regression in time

$$\mathcal{L}_{\text{data}}(\theta) = \frac{1}{12}\sum_{i=1}^{12}\left(x_\theta(t_i) - x(t_i)\right)^2$$

data: 12 pairs $(t_i, x(t_i))$, t_i evenly spaced on $[0, 4]$, exact solution, no noise; $x_\theta(t)$: network with input t; solution wanted up to $t = 10$

- an ordinary curve fit; the equation is not used
- largest error inside the window 2.6×10^{-2}, beyond it 1.2×10^{0}, as large as the oscillation
- the data alone carry no information beyond $t = 4$

Neural Network as a Trial Solution

Trial solution

$x(t) \approx x_\theta(t)$: a network with input t and output x; choose θ so that x_θ satisfies the equation

θ: all weights and biases of the network

- initial conditions built in, $x_\theta(t) = x_0 + v_0\, t + t^2 N_\theta(t)$ with a network N_θ (Lagaris *et al.*), or imposed by a penalty in the loss
- mesh-free: x_θ is defined at every t, not only at grid points
- physics-informed neural network (Raissi *et al.*): automatic differentiation, and measured data in the same loss

I. E. Lagaris, A. Likas, D. I. Fotiadis, IEEE TNN 9, 987 (1998); M. Raissi, P. Perdikaris, G. E. Karniadakis, J. Comput. Phys. 378, 686 (2019)

Residual of the Law

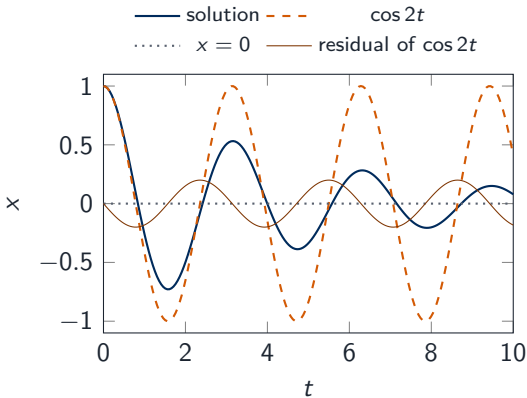
Residual: insert the network into the equation

$$r_\theta(t) := \frac{m_0\,\ddot{x}_\theta(t) + c\,\dot{x}_\theta(t) + k_s\,x_\theta(t)}{k_s}$$

$\dot{x}_\theta$, $\ddot{x}_\theta$: time derivatives of the network; division by k_s gives r_θ the unit of x

- $r_\theta(t)$ measures by how much x_θ violates the law at time t
- evaluated at collocation times t_j, chosen freely; no measurement is needed there
- $r = 0$ at every t together with the initial conditions: the solution, since it is unique

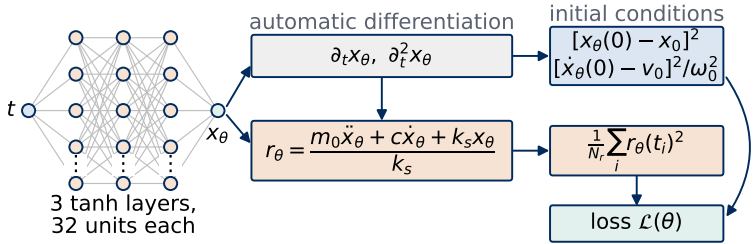
Zero Residual Needs the Initial Conditions



oscillator with $m_0 = 1$, $c = 0.4$, $k_s = 4$, $x(0) = 1$, $\dot{x}(0) = 0$

- $x = 0$: zero residual everywhere, wrong initial condition
- $x = \cos 2t$: right initial conditions, $r = -0.2 \sin 2t \neq 0$, friction missing
- zero residual at finitely many collocation times is weaker: a function may violate the law in between

Physics-Informed Neural Network (PINN)



Loss of the forward problem

$$\mathcal{L}(\theta) = \frac{1}{N_r}\sum_{j=1}^{N_r} r_\theta(t_j)^2 + w_{\mathrm{IC}}\Big([x_\theta(0) - x_0]^2 + \frac{[\dot{x}_\theta(0) - v_0]^2}{\omega_0^2}\Big)$$

network $x_\theta(t)$; automatic differentiation gives $\dot{x}_\theta$, $\ddot{x}_\theta$ and the residual r_θ at N_r collocation times t_j; w_{IC}: weight of the initial conditions; no solution data needed; k in the drawing: k_s

Derivatives by Automatic Differentiation

One hidden layer, written out

$$x_\theta(t) = \sum_j a_j \tanh(w_j t + b_j) + a_0$$

$$\dot{x}_\theta(t) = \sum_j a_j w_j (1 - \tanh^2(w_j t + b_j))$$

$$\ddot{x}_\theta(t) = -2 \sum_j a_j w_j^2 \tanh(w_j t + b_j)(1 - \tanh^2(w_j t + b_j))$$

$\theta = \{a_j, w_j, b_j, a_0\}$; exact functions of t, for any depth; tanh is smooth, a ReLU network has $\ddot{x}_\theta = 0$ almost everywhere

- `jax.grad` applies this chain rule automatically: no finite differences, no grid, no truncation error

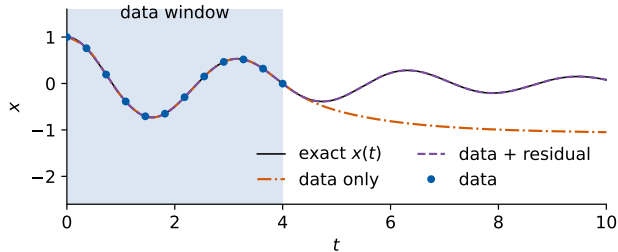
Data Plus the Law

The law as a soft constraint in the loss

$$\mathcal{L}(\theta) = \underbrace{\frac{1}{12}\sum_{i=1}^{12}\big(x_\theta(t_i) - x(t_i)\big)^2}_{\text{data}} + \underbrace{\frac{1}{N_r}\sum_{j=1}^{N_r} r_\theta(t_j)^2}_{\text{residual of the law}}$$

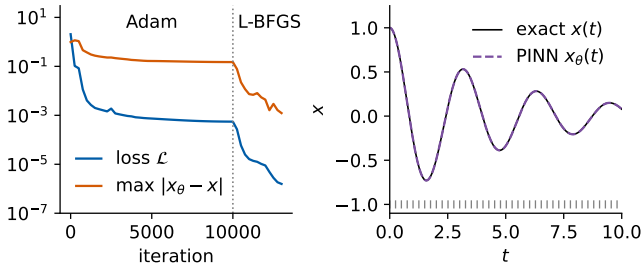
data: 12 pairs on $[0, 4]$; collocation times t_j: 200 points on $[0, 10]$, no target

- same data, same network, plus the residual; the law is penalised, not imposed
- beyond the data: error 1.2×10^{0} with data only, 3.0×10^{-4} with the law



shaded: data window

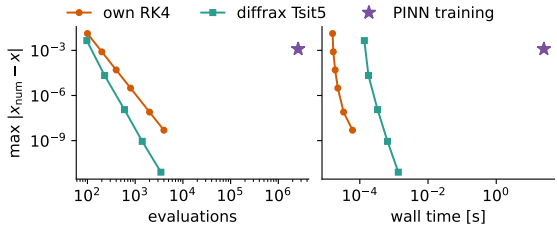
Training a PINN



left: loss $\mathcal{L}$ and largest error $\max_t |x_\theta - x|$ against the iteration; right: exact $x(t)$ and $x_\theta(t)$, ticks: collocation times

- Adam alone stalls at error 1.5×10^{-1}; then the limited-memory Broyden–Fletcher–Goldfarb–Shanno method (L-BFGS), a quasi-Newton method, reaches 1.2×10^{-3}
- 2209 parameters, about 25 s of training

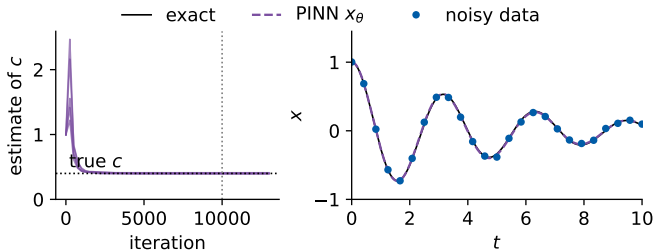
PINN against a Classical Solver



largest error against evaluations of the right-hand side (left) and wall time (right); RK4 with several steps Δt, adaptive Tsitouras 5(4) Runge–Kutta (Tsit5, diffrax), PINN training

- RK4, $\Delta t = 0.1$: error 5.0×10^{-5} with 400 evaluations
- PINN: 1.2×10^{-3} after 2.6×10^{6} evaluations of the residual
- known equation and initial conditions: the solver is more accurate with far fewer evaluations

Unknown Damping from Noisy Positions



c becomes a trainable parameter next to θ; left: its estimate during training for 5 noise realisations; right: noisy data and $x_\theta(t)$

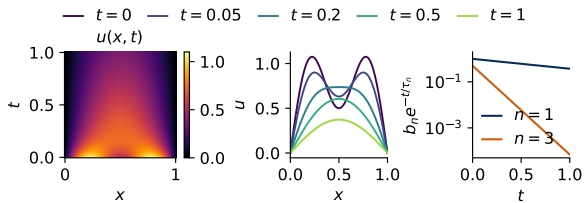
- PINN: $c = 0.401 \pm 0.006$; least-squares fit of the exact solution: 0.400 ± 0.004; true 0.4
- same task, same accuracy: the PINN finds an unknown parameter of a known equation

Diffusion in a Rod

Heat equation

$$\partial_t u = D\,\partial_x^2 u, \qquad u(0,t) = u(1,t) = 0, \qquad u(x,0) = \sin \pi x + \tfrac{1}{2} \sin 3\pi x$$

$u(x,t)$: temperature, ends held at zero; $D = 0.1$: thermal diffusivity



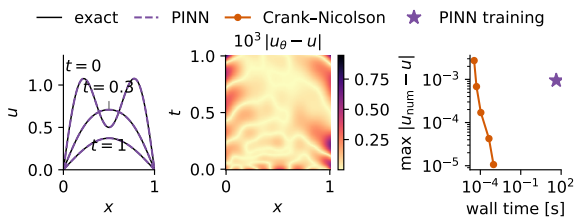
left: $u(x,t)$; middle: profiles at several times; right: amplitudes of the modes $n = 1, 3$, logarithmic axis

- exact: $u = e^{-D\pi^2 t} \sin \pi x + \frac{1}{2} e^{-9D\pi^2 t} \sin 3\pi x$; mode n decays in $\tau_n = 1/(Dn^2\pi^2)$: $\tau_1 = 1.013$, $\tau_3 = 0.113$

PINN for the Heat Equation

Loss

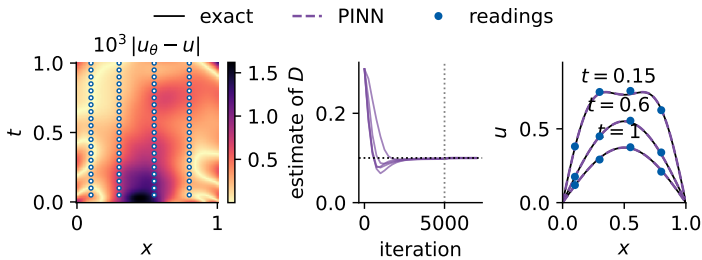
$$\mathcal{L} = \frac{1}{N_r}\sum_j \left(\partial_t u_\theta - D\,\partial_x^2 u_\theta\right)^2\Big|_{(x_j,t_j)} + \frac{1}{N_0}\sum_i \left(u_\theta(x_i,0) - u(x_i,0)\right)^2 + \text{boundary terms}$$



left: profiles; middle: $|u_\theta - u|$ over (x, t); right: largest error against wall time

- PINN $u_\theta(x,t)$, 2241 parameters, 1681 collocation points: largest error 9.5×10^{-4} after 44 s
- Crank–Nicolson, 20 points: 6.8×10^{-4} in 5.2×10^{-5} s

Diffusivity from Four Thermometers



4 thermometers, 20 readings each, noise $\sigma = 0.01$; D trainable, start 0.3; loss: readings + residual + initial and boundary terms

- PINN: $D = 0.09983 \pm 0.00057$; least squares through Crank–Nicolson: 0.09978 ± 0.00052; true 0.1
- the same network gives the field $u(x, t)$ between the thermometers, error 1.6×10^{-3}; initial profile known here, so the classical fit does as well

Particle in a Harmonic Trap

Schrödinger equation, $\hbar = m_0 = \omega = 1$

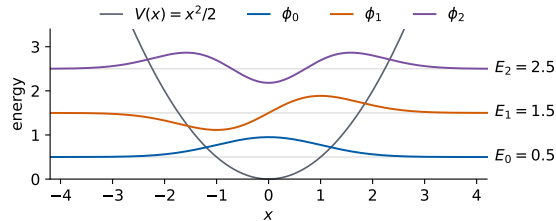
$$i\,\partial_t \psi = -\tfrac{1}{2}\partial_x^2 \psi + \tfrac{1}{2} x^2 \psi$$

eigenstates $\psi_n(x)$, energies $E_n = n + \frac{1}{2}$

Superposition of the two lowest states

$$\psi(x,t) = \tfrac{1}{\sqrt{2}}\left(e^{-iE_0 t}\psi_0 + e^{-iE_1 t}\psi_1\right)$$

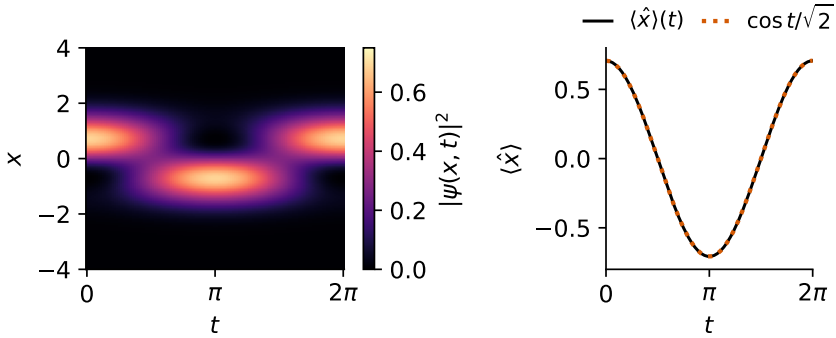
$|\psi|^2$ oscillates with period 2π;
$\langle \hat{x} \rangle(t) = \cos t/\sqrt{2}$; norm $\int |\psi|^2 dx = 1$



$V(x) = x^2/2$; lines: E_n; curves: ψ_0, ψ_1, ψ_2, written ϕ_n in the drawing, scaled by 0.6 and shifted by E_n

D. J. Griffiths, D. F. Schroeter, *Introduction to Quantum Mechanics*, 3rd ed. (2018)

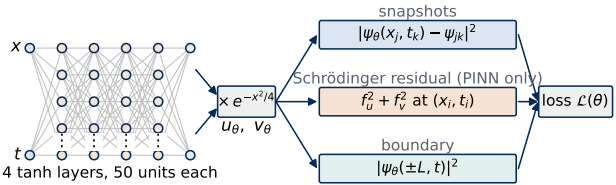
Exact Evolution in the Trap



left: $|\psi(x,t)|^2$, time horizontal, position vertical; right: $\langle\hat{x}\rangle(t)$ and $\cos t/\sqrt{2}$

- the wave packet swings with period 2π and returns to its initial form

Evolution from Three Snapshots



Task

snapshots $\psi(x_j, t_k)$, $t_k = 0, 0.5, 1$, 128 points, noise $\sigma = 0.01$; predict ψ for $0 \le t \le 2\pi$, inside the collocation domain

$\psi_\theta = u_\theta + iv_\theta = e^{-x^2/4}(n_\theta^{\mathrm{re}} + i\, n_\theta^{\mathrm{im}})$, network outputs n_θ^{re}, n_θ^{im} with inputs (x, t); residual $f_u + if_v = i\partial_t\psi_\theta + \frac{1}{2}\partial_x^2\psi_\theta - \frac{1}{2}x^2\psi_\theta$

- synthetic snapshots with the global phase of the exact solution (tomography leaves it undetermined); PINN: snapshot, boundary and residual terms (3072 points)

Two Real Residuals and the Norm

Residual of $\psi_\theta = u_\theta + iv_\theta$, real V

$$r_\theta := i\,\partial_t\psi_\theta - \hat{H}\psi_\theta, \qquad f_u = \operatorname{Re} r_\theta = -\partial_t v_\theta - \hat{H}u_\theta, \qquad f_v = \operatorname{Im} r_\theta = \partial_t u_\theta - \hat{H}v_\theta$$

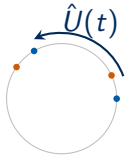
Norm $N_\theta(t) := \int |\psi_\theta|^2 dx$

exact: $\partial_t|\psi|^2 = -\partial_x j$, $j = \operatorname{Im}(\psi^*\partial_x\psi)$, so $dN/dt = 0$

network: $dN_\theta/dt = 2\operatorname{Im}\langle\psi_\theta|r_\theta\rangle$

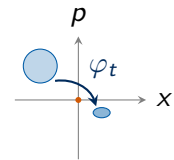
- zero residual for every solution, including every $\psi_n e^{-iE_n t}$ and $\psi = 0$
- the snapshots select the solution, the residual moves it in time
- norm drift of the network $\Rightarrow$ residual $\neq 0$; constant norm proves less

Unitary Evolution and the Classical Flow Map



two states keep their distance

$$\hat{U}(t) = e^{-i\hat{H}t}, \quad \|\hat{U}\chi - \hat{U}\chi'\| = \|\chi - \chi'\|$$



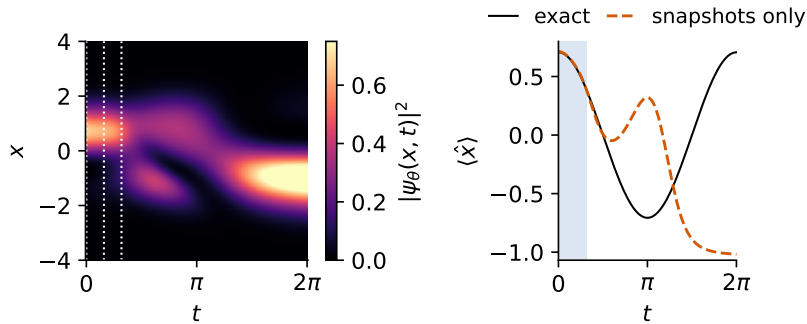
damped oscillator: areas shrink

$$d\mathcal{V}/dt = \int \nabla \cdot \mathbf{F}\, d\mathbf{x}, \quad \nabla \cdot \mathbf{F} = -c/m_0$$

$\mathbf{F}$: vector field of $\dot{x} = p/m_0$,
$\dot{p} = -k_s x - c\,p/m_0$; mass m_0, friction c;
Hamiltonian flow: $\nabla \cdot \mathbf{F} = 0$ (Liouville)

- Schrödinger = linear Hamiltonian flow: $\partial_t u = \hat{H}v$, $\partial_t v = -\hat{H}u$
- exact law: initial error carried with unchanged size; dissipation forgets it
- network: $\|\psi_\theta(t) - \psi(t)\| \leq \|\psi_\theta(0) - \psi(0)\| + \int_0^t \|r_\theta\|\, dt'$

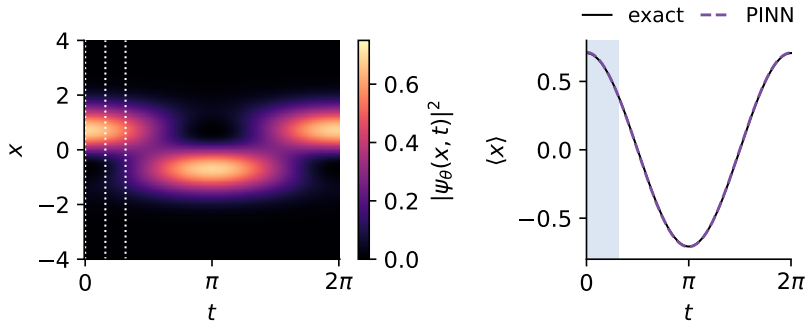
Snapshots Only



left: $|\psi_\theta(x,t)|^2$, dotted: snapshot times; right: $\langle\hat{x}\rangle(t)$ of the model and exact, shaded: snapshot window

- error 1.3×10^{-2} for $t \leq 1$, 1.2×10^{0} after: the packet drifts and does not return
- norm between 0.719 and 1.652: probability is lost

PINN with the Schrödinger Residual

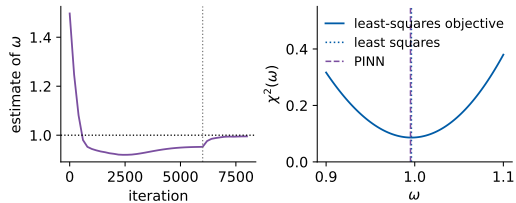


as before for the PINN; error: largest $|\psi_\theta - \psi|$ over x at each time

- error 6.7×10^{-3} for $t \leq 1$, 4.0×10^{-3} after: the oscillation continues over a full period
- norm between 0.999 and 1.002; training 73 s against 15 s

PINN against Classical Quantum Dynamics

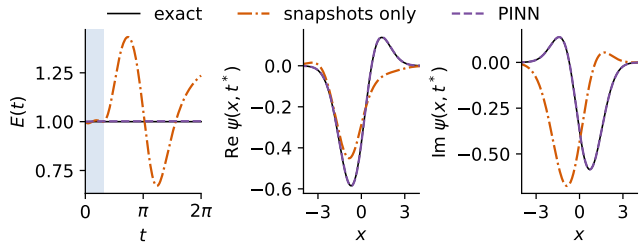
- split-step Fourier from exact $\psi(x, 0)$: 8.4×10^{-6} in 9.1×10^{-3} s; PINN 4.0×10^{-3} in 73 s
- eigenstate expansion of the same three snapshots, the fair reference: 6.1×10^{-3}; PINN matches it
- unknown ω (true 1): PINN 0.9952 (one run); least squares through split-step 1.0000 ± 0.0033 (20 noise draws)



all methods use the known Hamiltonian; right: ω during training and least-squares objective $Q(\omega)$ (χ^2 in the drawing)

J. A. Fleck, J. R. Morris, M. D. Feit, Appl. Phys. 10, 129 (1976)

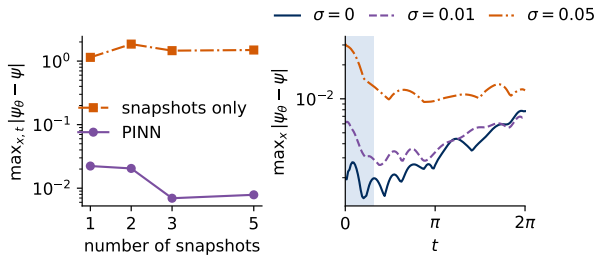
Energy and Phase



$E(t) = \langle\psi_\theta|\hat{H}|\psi_\theta\rangle / \langle\psi_\theta|\psi_\theta\rangle$, exact $E = 1$; middle and right: $\mathrm{Re}\,\psi$, $\mathrm{Im}\,\psi$ at $t^* = 3\pi/2$

- PINN: energy conserved to 2.3×10^{-3}, real and imaginary parts correct, error 2.8×10^{-3} at t^*
- snapshots only: E wanders between 0.67 and 1.43
- conservation is not in the loss; the residual is a soft penalty: without the envelope the PINN's norm falls to 0.545

Number of Snapshots



left: largest error over $0 \leq t \leq 2\pi$ against the number of snapshots, spaced by 0.5, reduced training budget; right: PINN error over time for three noise levels

- snapshots only: error of order one for 1, 2, 3 and 5 snapshots
- PINN: 2.2×10^{-2} with one snapshot, 7.0×10^{-3} with three; noise $\sigma = 0.05$ gives 3.1×10^{-2}

Double Well

Double well, $\hbar = m_0 = 1$

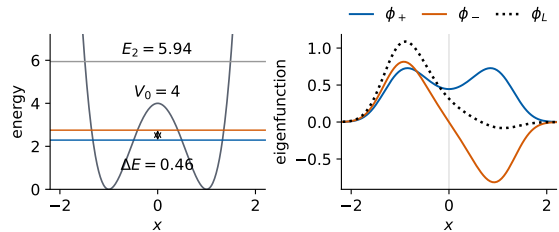
$$V(x) = V_0(x^2/a^2 - 1)^2, \quad V_0 = 4,\ a = 1$$

minima at $x = \pm a$, barrier V_0 at $x = 0$; lowest pair $\psi_\pm$, splitting $\Delta E = E_1 - E_0 = 0.462$, below the barrier

Left-localised state

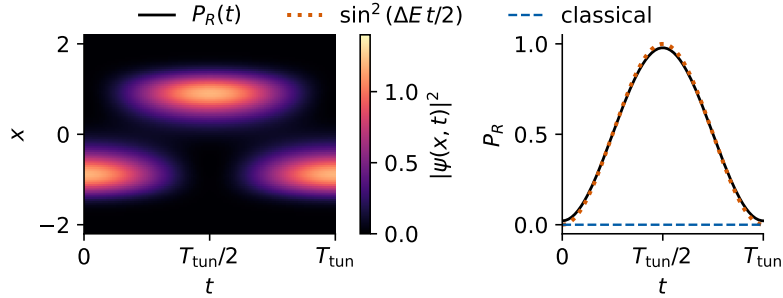
$$\psi_L = (\psi_+ + \psi_-)/\sqrt{2}$$

ψ_+ even, no node; ψ_- odd, node at $x = 0$



left: $V(x)$, levels E_0, E_1, E_2, splitting ΔE; right: ψ_+, ψ_-, ψ_L, written $\phi_\pm$, ϕ_L in the drawing

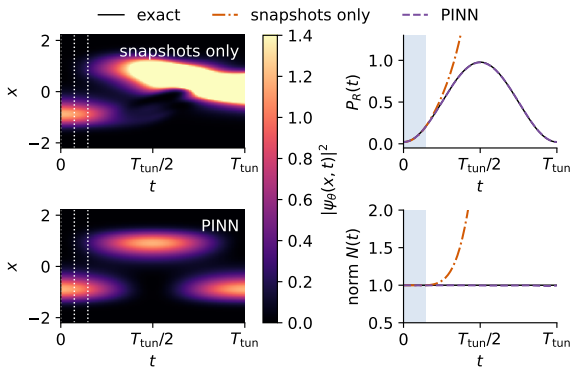
Tunnelling of the Left-Localised State



start $\psi(x,0) = \psi_L$; left: $|\psi(x,t)|^2$; right: $P_R(t) = \int_{x>0} |\psi|^2 dx$, the two-level formula $\sin^2(\Delta E\, t/2)$ and the classical value 0

- tunnelling period $T_{\mathrm{tun}} = 2\pi/\Delta E = 13.587$; a classical particle with this energy stays in the left well

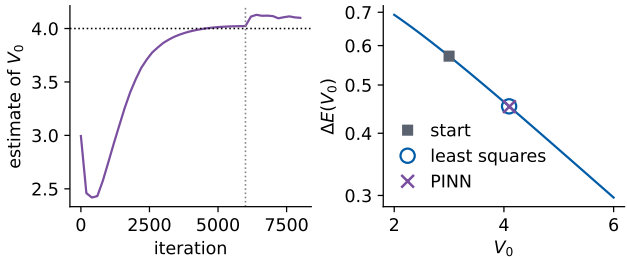
Tunnelling from Early Snapshots



snapshots $t = 0, 1, 2$ (first 14.7 % of T_{tun}); left: densities, snapshots only (top), PINN (bottom), dotted: snapshot times; right: $P_R(t)$, norm $N(t)$, shaded: snapshot window

- snapshots only: P_R exceeds 1, norm grows to 7.601
- PINN: tunnels and returns, error 7.9×10^{-3}; fitted period 13.60, exact 13.587
- used: envelope e^{-x^2}, $V - \bar{E}$, $\bar{E} = (E_0{+}E_1)/2$; with $e^{-x^2/4}$ or V: decay to $\psi = 0$

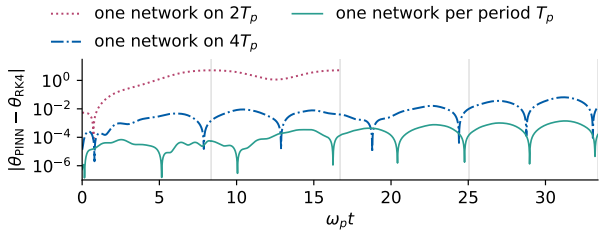
Unknown Barrier Height



V_0 trainable, start 3; right: $\Delta E(V_0)$ from the spectrum on a logarithmic axis

- PINN (one run): $V_0 = 4.100$; least squares through split-step: 4.094, over 20 noise draws 3.973 ± 0.079; true 4
- energy zero ε, residual with $V - \varepsilon$, learned too: 2.555, true $\bar{E} = 2.521$
- ΔE decreases with V_0 ($d \ln \Delta E / dV_0 = -0.22$ at $V_0 = 4$), so the tunnelling period carries information on V_0

PINN over Several Pendulum Periods



$\ddot{\theta} = -\omega_p^2 \sin\theta$, $\theta(0) = 2$, at rest; $|\theta_{\mathrm{PINN}} - \theta_{\mathrm{RK4}}|$ against $\omega_p t$, first of three initialisations; grey lines: period boundaries T_p

- one network on $2T_p$: all 3 of 3 seeds leave the solution at $t \approx 1.95$ and settle near $\theta = -\pi$, error 5.1×10^0 at loss 3.1×10^{-3} (one period: $\leq 6.4 \times 10^{-8}$); on $4T_p$: 2 of 3 succeed (6.5×10^{-2}, 5.2×10^{-2})
- one network per period, started from the end of the previous one: error 1.4×10^{-3} after four periods

Strengths and Limits of PINNs

Strengths

- inverse problems: unknown parameters of a known law, here the damping c
- sparse, noisy data combined with the law in one loss
- mesh-free: irregular domains and many dimensions

Limits

- forward problems: far slower than a classical solver
- hard optimisation: loss terms of very different size pull the weights in conflicting directions
- long times: wrong functions with a small residual, here near $\theta = -\pi$; one network per period avoided it in these runs

S. Wang, Y. Teng, P. Perdikaris, SIAM J. Sci. Comput. 43, A3055 (2021); A. S. Krishnapriyan *et al.*, NeurIPS (2021), arXiv:2109.01050

10
Hybrid Models

JAGIELLONIAN UNIVERSITY
IN KRAKOW

One Pendulum, Three Models

$$\ddot{\theta} = \underbrace{-\omega_p^2 \sin\theta}_{\text{gravity: known}} + \underbrace{F(\dot{\theta})}_{\text{friction: unknown}}$$

1 $\ddot{\theta} = -\omega_p^2 \sin\theta$ — gravity only: swings forever

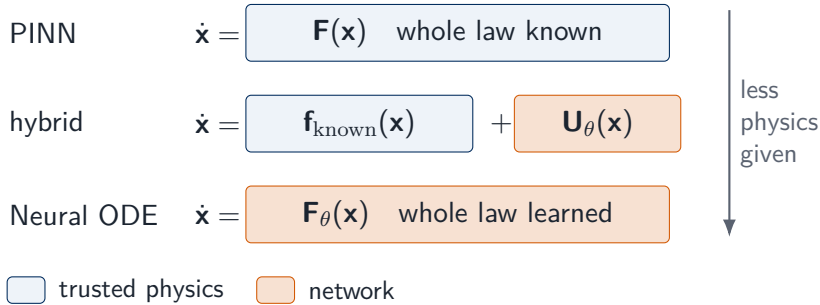
2 $(\dot{\theta}, \ddot{\theta}) = \mathbf{F}_\theta(\theta, \dot{\theta})$ — Neural ODE: learns everything

3 $\ddot{\theta} = -\omega_p^2 \sin\theta + U_\theta(\theta, \dot{\theta})$ — **hybrid**: network learns only friction

formula from U_θ: $U \approx -0.1497\, \dot{\theta}|\dot{\theta}|$, air drag

- hybrid: the physics we know plus a network for the missing piece only
- then the learned piece is turned back into a formula, valid beyond the data as far as the true law has its form

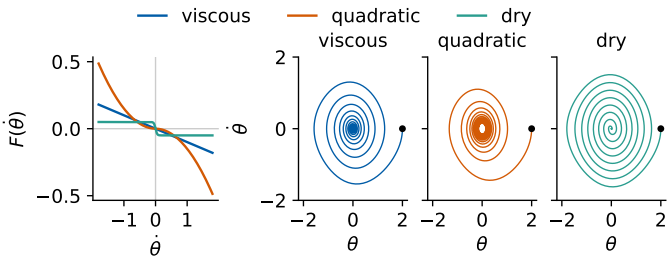
Known Terms and a Learned Term



- hybrid model, or universal differential equation: network only for the uncertain term
- trained through the solver, like a Neural ODE
- then sparse regression turns $\mathbf{U}_\theta$ into a short formula

C. Rackauckas *et al.*, arXiv:2001.04385; S. L. Brunton, J. L. Proctor, J. N. Kutz, PNAS 113, 3932 (2016)

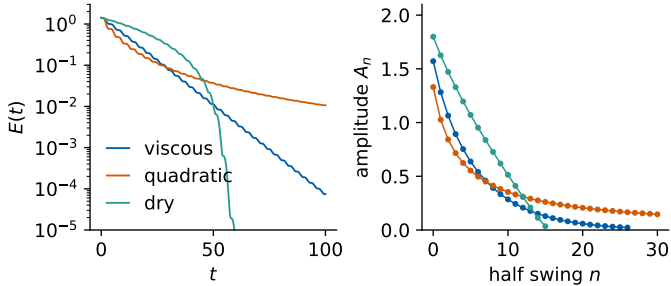
Pendulum with Unknown Friction



$$\ddot\theta = \underbrace{-\omega_p^2 \sin\theta}_{\text{gravity, known}} + \underbrace{F(\dot\theta)}_{\text{friction, unknown}}$$

viscous $F = -\gamma\dot\theta$ (slow motion in a fluid); quadratic $F = -c\,\dot\theta|\dot\theta|$ (air drag); dry $F = -\mu\tanh(\dot\theta/\varepsilon)$ (Coulomb, $\varepsilon = 0.05$); $\omega_p = 1$

Energy Loss per Swing



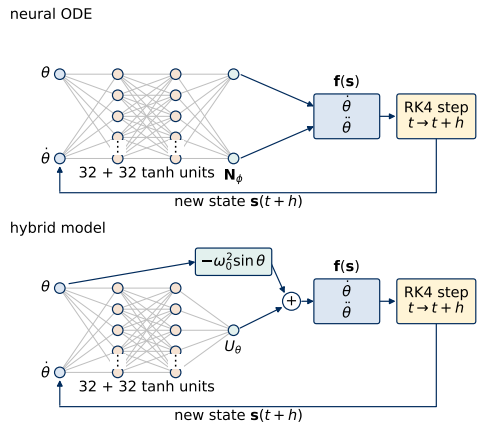
$E := \frac{1}{2}\dot{\theta}^2 + \omega_p^2(1-\cos\theta)$; A_n: amplitude at the n-th turning point; small-amplitude slopes fitted $-0.159, 0.201, -0.103$, predicted $-0.157, 0.200, -0.100$

- viscous: $\ln A_n$ linear; quadratic: $1/A_n$ linear; dry: A_n linear
- each law leaves its own signature in the decay

Three Models of the Pendulum

- gravity only: $\ddot{\theta} = -\omega_p^2 \sin\theta$, no parameters
- Neural ODE: $\dot{\mathbf{x}} = \mathbf{F}_\theta(\mathbf{x})$, $\mathbf{x} = (\theta, \dot{\theta})$, 1218 parameters
- hybrid: $\ddot{\theta} = -\omega_p^2 \sin\theta + U_\theta(\theta, \dot{\theta})$, 1185 parameters

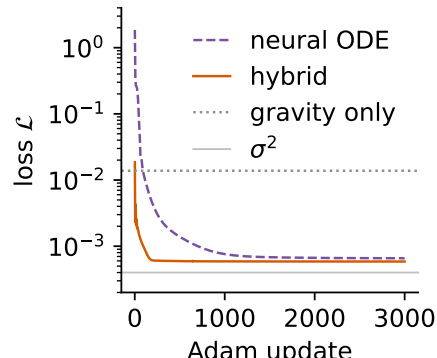
data: one swing with quadratic drag, $c = 0.15$ hidden; 201 noisy states on $0 \le t \le 20$, $\sigma = 0.02$; drawing: state $\mathbf{s}$, field $\mathbf{f}(\mathbf{s})$, Neural ODE network N, RK4 step h



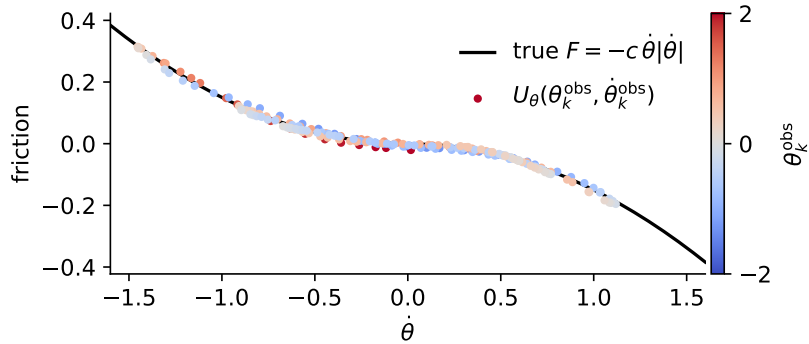
Training through the Solver

- multiple shooting: integrate from each measured state over 30 samples (half a period), compare with the next measurements
- short segments keep the phase error small
- final loss: hybrid 0.00059, Neural ODE 0.00066, gravity only 0.0138

noise floor $\sigma^2 = 4 \times 10^{-4}$ per component; the noise of each segment's measured start adds to it



Learned Friction



dots: network output U_θ at the measured states, colour: angle; line: true $F = -c\,\dot\theta|\dot\theta|$

- relative error $\mathrm{rms}(U_\theta - F)/\mathrm{rms}(F) = 0.076$ on the measured states, no friction formula given
- a function of velocity, not yet a law

Sparse Regression

Network output as a short sum of candidate terms

$$U_\theta(\mathbf{s}_k^{\mathrm{obs}}) \approx \sum_j w_j \, \Theta_j(\mathbf{s}_k^{\mathrm{obs}})$$

$$\Theta = \{1,\ \sin\theta,\ \theta^2,\ \theta\dot{\theta},\ \dot{\theta},\ \dot{\theta}|\dot{\theta}|,\ \tanh(\dot{\theta}/\varepsilon),\ \dot{\theta}^3\}$$

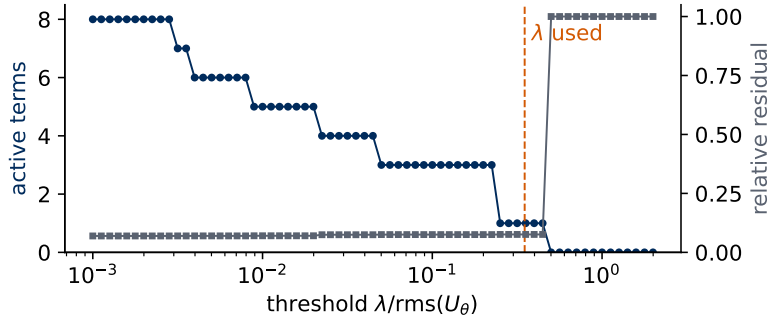
Θ_j: library of candidate terms, each column divided by its rms over the data; w_j: coefficients; $\mathbf{s}_k^{\mathrm{obs}}$: measured states

1. least-squares fit of all w_j
2. set every $|w_j| < \vartheta$ to zero
3. refit the remaining terms, repeat until nothing changes

sequentially thresholded least squares (STLSQ); ridge penalty $0.01\, n\|\mathbf{w}\|^2$ inside the iteration against cancelling collinear velocity terms, final set refitted without it

S. L. Brunton, J. L. Proctor, J. N. Kutz, PNAS 113, 3932 (2016)

One Friction Term Survives



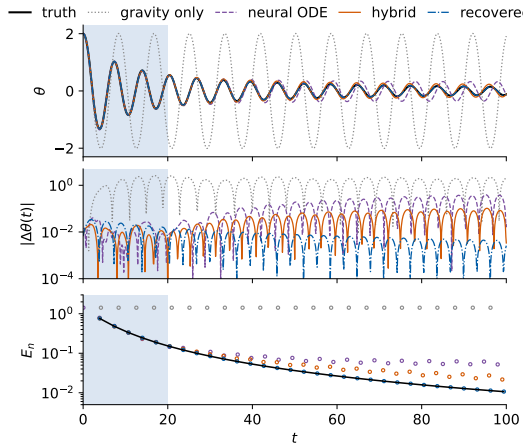
number of active terms (left axis) and relative residual $\mathrm{rms}(U_\theta - \Theta\mathbf{w})/\mathrm{rms}(U_\theta)$ (right axis) against $\vartheta/\mathrm{rms}(U_\theta)$ (λ in the drawing); dashed: $\vartheta = 0.35\,\mathrm{rms}(U_\theta)$, used

- one term $-c\,\dot\theta|\dot\theta|$, $c = 0.1497$ (true 0.15), only for ϑ from 0.251 to 0.45 times $\mathrm{rms}(U_\theta)$, less than a factor 2
- same regression on finite differences of the raw data: no term, noise dominates

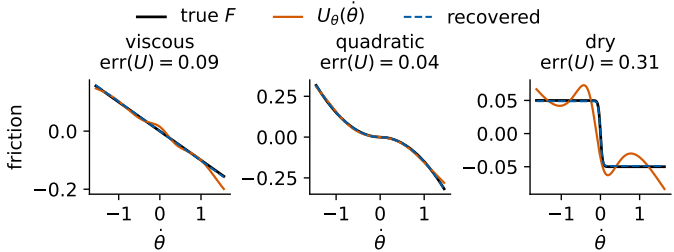
Pendulum Forecast to Five Times the Data Window

integrated from the first measured state to $t = 100$; shaded: data; recovered: gravity plus the STLSQ formula; E_n: energy at the turning points

- RMS $\Delta\theta$ for $20 < t \leq 100$:
  gravity only 1.479
  Neural ODE 0.176
  hybrid 0.047
  recovered 0.006
- recovered equation:
  $E(100) = 0.0107$, true 0.0107



Three Friction Laws Recovered

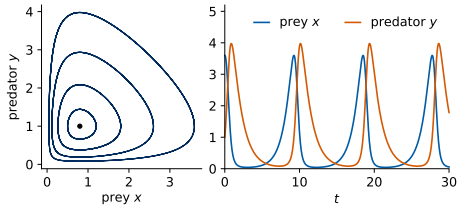


same pipeline, new data for each law; network input $(\theta, \dot\theta)$ or $\dot\theta$ only (figure)

law	true	recovered, $U_\theta(\theta, \dot\theta)$	recovered, $U_\theta(\dot\theta)$
viscous γ	0.1	0.1008	0.1017
quadratic c	0.15	0.1497	0.1505
dry μ	0.05	0.0488	0.0489

correct single term in all six cases; the library contains the true ε of the dry law

Lotka–Volterra Equations



Predator and prey

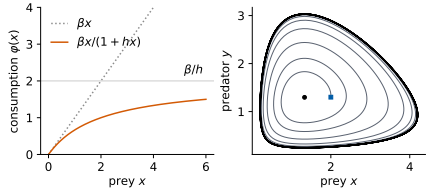
$$\dot{x} = \alpha x - \beta xy, \qquad \dot{y} = -\delta y + \gamma xy$$

x: prey, y: predators; α: prey birth rate; δ: predator death rate; βxy, γxy: encounters; $\alpha = \beta = \gamma = 1$, $\delta = 0.8$; left: orbits in (x, y); right: populations against t

- conserved $\mathcal{H} = \gamma x - \delta \ln x + \beta y - \alpha \ln y$: closed orbits around $(\delta/\gamma, \alpha/\beta)$, amplitude fixed by the start

A. J. Lotka, PNAS 6, 410 (1920); V. Volterra, Nature 118, 558 (1926)

Saturating Predator and Limited Prey



Rosenzweig–MacArthur model

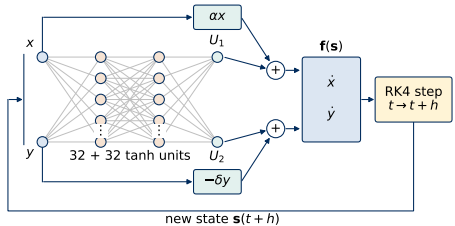
$$\dot{x} = \alpha x - \frac{\alpha}{K} x^2 - \frac{\beta x y}{1 + h x}, \qquad \dot{y} = -\delta y + \frac{\gamma x y}{1 + h x}$$

$\beta x/(1 + hx)$: prey eaten per predator, saturates through the handling time; K: carrying capacity; $h = 0.5$, $K = 6$; generates the data, hidden from the models; left: response against x; right: orbits onto the limit cycle

- unstable focus, every start ends on one limit cycle, period 12.53

C. S. Holling, Can. Entomol. 91, 385 (1959); M. L. Rosenzweig, R. H. MacArthur, Am. Nat. 97, 209 (1963)

Hybrid Predator–Prey Model



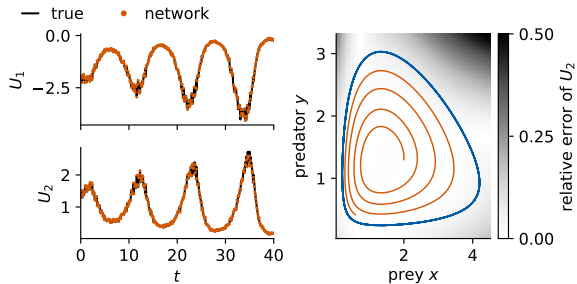
Known single-species terms, learned interaction

$$\dot{x} = \alpha x + U_1(x, y), \qquad \dot{y} = -\delta y + U_2(x, y)$$

$\alpha = 1$, $\delta = 0.8$ from separate experiments; truth: $U_1 = -\frac{\alpha}{K}x^2 - \frac{\beta xy}{1+hx}$, $U_2 = \frac{\gamma xy}{1+hx}$, hidden from the model

- data: 401 counts on $0 \le t \le 40$, 5 % noise, still spiralling out: prey up to 3.45, on the cycle up to 4.20

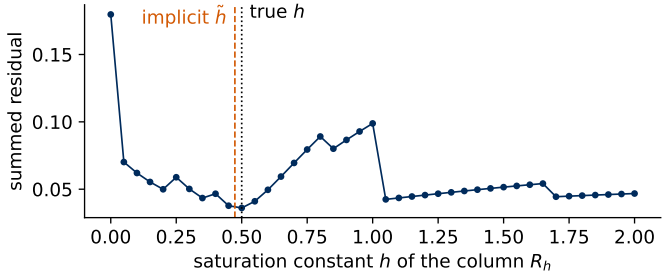
Learned Interaction



left: U_1, U_2 along the measured trajectory; right: relative error of U_2, blue: limit cycle, orange: data

- error on the data: 0.015 (U_1), 0.022 (U_2); on the limit cycle up to 0.08
- accurate where the data are; the upper right corner is never visited

Rational Term by a Linear Fit



residual of the linear fit against the saturation constant h

- polynomials only: a truncated Taylor series of $1/(1+hx)$, no readable law
- multiply by the denominator: $U_2 = \gamma\, xy - h\,(x\, U_2)$, linear in γ, h
- fitted $\tilde{h} = 0.474$, true 0.5

N. M. Mangan, S. L. Brunton, J. L. Proctor, J. N. Kutz, IEEE TMBMC 2, 52 (2016)

Recovered Interaction

$$U_1 = a_1 x^2 + a_2 R, \qquad U_2 = a_3 R, \qquad R := \frac{xy}{1 + hx}$$

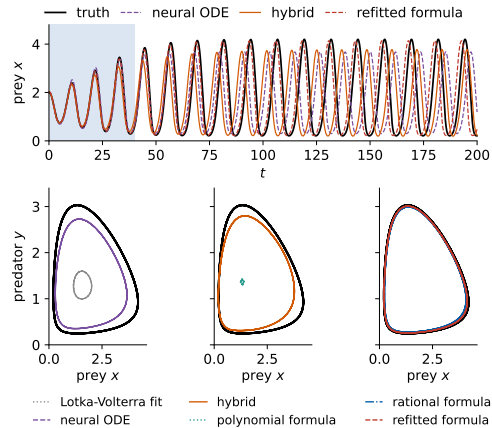
	h	a_1	a_2	a_3
STLSQ with $\tilde{h}$	0.474	-0.165	-0.973	0.970
refit through the solver	0.492	-0.167	-0.991	0.991
true	0.5	$-1/6$	-1	1

- STLSQ keeps exactly the true terms: logistic x^2 and saturating predation
- refit of the short formula through the solver: h and all coefficients within 1.6 % of the truth

Predator–Prey Forecast

model	period	prey max
truth	12.53	4.20
Lotka–Volterra fit	11.40	2.01
Neural ODE	11.75	3.68
hybrid	11.70	3.78
polynomial formula	8.90	1.41
refitted formula	12.40	4.14

- networks: cycle too small, too fast
- refitted rational formula closest; polynomial formula fails



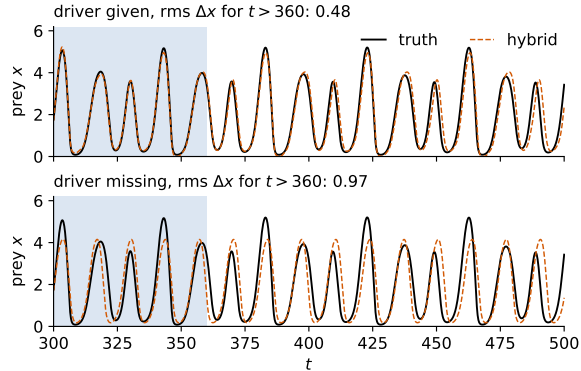
integrated to $t = 200$; top: prey against t, shaded: data; bottom: late cycles in (x, y)

Seasonal Driver

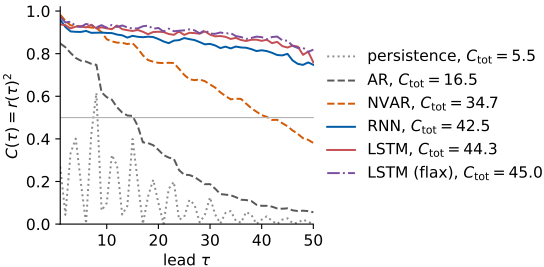
$$\alpha(t) = \alpha_0(1 + a_s \sin \omega t)$$

rain-driven prey growth, $\alpha_0 = 1$, $a_s = 0.2$, season $2\pi/\omega = 20$; $\alpha_0 a_s \sin(\omega t)\, x$ belongs to the unknown U_1; right: prey, driven truth and hybrid model, top with the driver as input, bottom without

- driver as input: three-season pattern; STLSQ finds $0.193\, x \sin \omega t$, true $0.2\, x \sin \omega t$
- driver missing: one closed cycle, error 0.754 already on the data



The Laser Record Revisited



capacity $C(\tau)$ on the Santa Fe test block against the horizon τ, written lead; totals C_{tot} in the legend; LSTM (flax): the same LSTM built from the flax library cell

The law behind the record

- surrogates reject linear noise: nearest-neighbour error 0.208, surrogates ≥ 0.670
- $m = 3$, $D_2 \approx 2.46$; $\lambda_1 = 0.029$–0.065 per sample
- Maxwell–Bloch = Lorenz form, three variables

Forecast range

- C_{tot}: persistence 5.5, AR 16.5, NVAR 34.7, LSTM 44.3

Models That Read the Record

- AR, MA, NVAR: linear readouts of fixed features of the window; delay DMD reduces to AR, the Koopman approximation with the raw past values as dictionary
- MLP, RNN, GRU, LSTM learn the features; gated cells pass values and gradients along a memory path
- recurrent networks scored highest on the laser and the chaotic records (MG 35: MLP and recurrent networks all within 0.5 of 50), for these records, protocol and model sizes

Models with Physics Inside

- Neural ODE: a network as the vector field, trained through the solver; reliable where the data are; its divergence tells dissipative from area preserving (in 2D: Hamiltonian)
- PINN: the residual of a known law in the loss; suited to unknown parameters and sparse data, slower than a solver for a plain forward problem
- hybrid model: known terms kept, a network for the missing one, then a formula by sparse regression, valid as far as the library contains the true form; a driver outside the state must be an input

Thank you for your attention